1 Principles of Monte Carlo

1.1 Introduction

Monte Carlo integration

Consider estimating $\alpha = \int_0^1 f(x)\,dx$. If $U \sim \mathrm{Unif}(0,1)$, then $\alpha = \mathbb{E}[f(U)]$. Given independent samples $U_1,\dots,U_n \sim \mathrm{Unif}(0,1)$, the Monte Carlo estimator is

$$\hat\alpha_n = \frac1n \sum_{i=1}^n f(U_i).$$

By the strong law of large numbers, $\hat\alpha_n \to \alpha$ a.s. as $n\to\infty$. By the central limit theorem,

$$\hat\alpha_n - \alpha \approx N(0,\sigma_f^2/n),$$

where $\sigma_f^2 = \mathrm{var}(f(U))$. Thus the standard error is $\sigma_f/\sqrt n$. Halving the standard error requires increasing the number of samples by a factor of 4.

Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is $O(n^{-2})$. The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

1.2 First Example: European Call Option

Risk-neutral valuation

Let $S(t)$ be the stock price at time t. A European call option with strike K and maturity T has payoff $(S(T)-K)^+ = \max\{0, S(T)-K\}$. Under the risk-neutral measure $\mathbb{Q}$, the time-0 price is

$$V_0 = \mathbb{E}_0^{\mathbb{Q}}\left[e^{-rT}(S(T)-K)^+\right],$$

where r is the continuously compounded risk-free rate.

Stock-price dynamics

Under $\mathbb Q$, geometric Brownian motion satisfies $\frac{dS(t)}{S(t)} = r\,dt + \sigma\,dW(t)$, where $W(t)$ is a standard Brownian motion. The exact solution at time T is

$$S(T) = S(0)\exp\left(\left(r-\tfrac12\sigma^2\right)T + \sigma W(T)\right).$$

Since $W(T) = \sqrt T Z$ with $Z\sim N(0,1)$,

$$S(T) = S(0)\exp\left(\left(r-\tfrac12\sigma^2\right)T+\sigma\sqrt T Z\right).$$

Simulation algorithm

For $i=1,\dots,n$: generate $Z_i\sim N(0,1)$, then set

$$S_i(T) = S(0)\exp\left(\left(r-\tfrac12\sigma^2\right)T+\sigma\sqrt T Z_i\right),$$

$$C_i = e^{-rT}(S_i(T)-K)^+.$$

The Monte Carlo estimator is $\hat C_n = \frac1n\sum_{i=1}^n C_i$. It is unbiased, $\mathbb E[\hat C_n]=C$, and strongly consistent, $\hat C_n\to C$ with probability 1.

Confidence interval

The sample standard deviation of the simulated discounted payoffs is

$$s_C = \sqrt{\frac{1}{n-1}\sum_{i=1}^n (C_i-\hat C_n)^2}.$$

An asymptotic $1-\delta$ confidence interval is

$$\hat C_n \pm z_{\delta/2}\frac{s_C}{\sqrt n}.$$

Since s_C is estimated, one may replace the normal quantile with the corresponding t-quantile with $n-1$ degrees of freedom.

Black-Scholes benchmark

The Black-Scholes value of a European call is

$$BS(S,\sigma,T,r,K) = S\Phi(d_1) - e^{-rT}K\Phi(d_2),$$

where

$$d_1 = \frac{\log(S/K)+(r+\frac12\sigma^2)T}{\sigma\sqrt T},$$

$$d_2 = \frac{\log(S/K)+(r-\frac12\sigma^2)T}{\sigma\sqrt T}.$$

1.3 Simulation Layers

Layered construction

Monte Carlo derivative pricing is built in layers:

$$U_i \to Z_i \to S_i(T) \text{ or } \{S_i(t_j)\}_{j=1}^m \to C_i \to \hat C_n,\ s_C.$$

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

1.4 Second Example: Asian Option

Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff $(\bar S-K)^+$, where

$$\bar S = \frac1m\sum_{j=1}^m S(t_j),$$

and $0=t_0<t_1<\cdots<t_m=T$. Its time-0 value is

$$\mathbb E_0^{\mathbb Q}\left[e^{-rT}(\bar S-K)^+\right].$$

To estimate this value, simulate the path values $S(t_1),\dots,S(t_m)$, compute the average $\bar S$, discount the payoff, and average over independent replications.

1.5 Efficiency of Simulation Estimators

Central limit theorem

Suppose $\hat C_n = \frac1n\sum_{i=1}^n C_i$, where C_i are i.i.d., $\mathbb E_0^{\mathbb Q}[C_i]=C$, and $\mathrm{var}(C_i)=\sigma_C^2<\infty$. Then

$$\frac{\hat C_n - C}{\sigma_C/\sqrt n}\Rightarrow N(0,1),$$

or equivalently

$$\sqrt n(\hat C_n - C)\Rightarrow N(0,\sigma_C^2).$$

Hence $\hat C_n - C\approx N(0,\sigma_C^2/n)$. For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

1.6 Bias

General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite n, but become asymptotically unbiased as computational effort increases.

Ratio estimator

For i.i.d. samples (X_i,Y_i), the estimator $\bar X/\bar Y$ for $\mathbb E[X]/\mathbb E[Y]$ is generally biased because

$$\mathbb E\left[\frac{\bar X}{\bar Y}\right]\ne\frac{\mathbb E[X]}{\mathbb E[Y]}.$$

However,

$$\frac{\bar X}{\bar Y}\to\frac{\mathbb E[X]}{\mathbb E[Y]}$$

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as $n\to\infty$.

Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

$$S((j+1)h) = S(jh)+rS(jh)h+\sigma S(jh)\sqrt h Z_{j+1},$$

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step h, at the cost of more transitions per path.

Payoff discretization error

For a continuously averaged Asian option,

$$\bar S = \frac1T\int_0^T S(u)\,du.$$

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a smaller simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payoff discretization error comes from approximating the payoff functional.

1.7 Mean Square Error

Bias-variance decomposition

For an estimator $\hat\alpha$ of α, the mean square error is

$$\mathrm{MSE}(\hat\alpha) = \mathbb E\left[(\hat\alpha-\alpha)^2\right].$$

It decomposes as

$$\mathrm{MSE}(\hat\alpha) = (\mathbb E[\hat\alpha]-\alpha)^2 + \mathbb E\left[(\hat\alpha-\mathbb E[\hat\alpha])^2\right],$$

or

$$\mathrm{MSE}(\hat\alpha) = \mathrm{Bias}^2(\hat\alpha)+\mathrm{Variance}(\hat\alpha).$$

Thus MSE balances bias and variance.

2 Generating Random Numbers and Random Variables

2.1 Random Number Generation

Uniform inputs

Monte Carlo simulation is driven by apparently random numbers. Ideally we want $U_1,U_2,\dots$ such that each $U_i\sim\mathrm{Unif}[0,1]$ and the U_i are mutually independent. Independence is crucial: U_i should not be predictable from $U_1,\dots,U_{i-1}$.

A pseudorandom number generator produces a finite deterministic sequence $u_1,\dots,u_K\in[0,1]$. A good generator makes short segments hard to distinguish from independent uniforms. Desirable properties: long period, reproducibility, speed, portability, and statistical randomness.

Linear congruential generator

A linear congruential generator is

$$x_{i+1} = ax_i \bmod m, \qquad u_{i+1}=x_{i+1}/m.$$

Here a is the multiplier, m the modulus, and $x_0\in\{1,\dots,m-1\}$ the seed. The modulo operation is

$$y \bmod m = y - \lfloor y/m\rfloor m.$$

Once a value repeats, the whole sequence repeats. The generator has full period if it produces all $m-1$ nonzero values before repeating. Large m alone is not enough; poor choices of a,m can produce short cycles.

2.2 General Sampling Methods

Uniform distribution

Assume independent uniforms satisfying $P(U_i\le u)=0$ if $u<0$, $P(U_i\le u)=u$ if $0\le u\le1$, and $P(U_i\le u)=1$ if $u>1$. Sampling methods transform these uniforms into variables with desired distributions.

2.3 Inverse Transform Method

General method

To sample X with CDF F, set

$$X=F^{-1}(U), \qquad U\sim\mathrm{Unif}[0,1].$$

If F is not strictly increasing, use the generalized inverse

$$F^{-1}(u)=\inf\{x:F(x)\ge u\}.$$

This chooses the smallest compatible x. Validity follows from

$$P(X\le x) = P(F^{-1}(U)\le x) = P(U\le F(x)) = F(x).$$

Flat sections of F have zero probability: if F is constant on $[a,b]$, then $P(a<X\le b)=F(b)-F(a)=0$.

Examples

Exponential with mean θ: $F(x)=1-e^{-x/\theta}$ for $x\ge0$, so

$$X=-\theta\log(1-U), \text{ equivalently } X=-\theta\log U.$$

Rayleigh example:

$$F(x)=1-e^{-2x(x-b)}, \qquad x\ge b,$$

so the inverse-transform sample is

$$X=\frac b2+\frac{\sqrt{b^2-2\log U}}{2}.$$

Discrete distribution: for outcomes $c_1<\cdots<c_n$ with probabilities p_i, define $q_0=0$ and

$$q_i=\sum_{j=1}^i p_j.$$

Generate U, find K with $q_{K-1}<U\le q_K$, and set $X=c_K$. Binary search can implement the lookup.

Conditional sampling

To sample X conditional on $a<X\le b$, with $F(a)<F(b)$, generate

$$V=F(a)+[F(b)-F(a)]U.$$

Then $F^{-1}(V)$ has the conditional distribution because

$$P(F^{-1}(V)\le x)=\frac{F(x)-F(a)}{F(b)-F(a)}.$$

2.4 Acceptance-Rejection Method

General method

To sample from a target density f on $\mathscr X$, choose a convenient proposal density g and a constant c such that

$$f(x)\le cg(x), \qquad x\in\mathscr X.$$

Generate $X\sim g$ and $U\sim\mathrm{Unif}[0,1]$. Accept X if

$$U\le\frac{f(X)}{cg(X)};$$

otherwise reject and repeat. If Y is the accepted value, then

$$P(Y\in A)=P(X\in A\mid U\le f(X)/(cg(X))).$$

The acceptance probability is

$$P(U\le f(X)/(cg(X)))=\int_{\mathscr X}\frac{f(x)}{cg(x)}g(x)\,dx=1/c.$$

Hence

$$P(Y\in A)=c\int_A\frac{f(x)}{cg(x)}g(x)\,dx=\int_A f(x)\,dx.$$

Thus Y has density f. The number of proposals until acceptance is geometric with mean c, so smaller c and tighter bounds are better.

Beta distribution

The beta density on $[0,1]$ is

$$f(x)=\frac{1}{B(\alpha_1,\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}.$$

Here

$$B(\alpha_1,\alpha_2)=\int_0^1 x^{\alpha_1-1}(1-x)^{\alpha_2-1}\,dx=\frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)},$$

and

$$\Gamma(z)=\int_0^\infty x^{z-1}e^{-x}\,dx, \qquad \mathrm{Re}(z)>0.$$

For integer $n>0$, $\Gamma(n)=(n-1)!$. If $\alpha_1,\alpha_2\ge1$ and at least one exceeds 1, the mode is

$$\frac{\alpha_1-1}{\alpha_1+\alpha_2-2}.$$

Let c be the maximum value of f and use $g(x)=1$ on $[0,1]$. Generate U_1,U_2 until $cU_2\le f(U_1)$, then return U_1.

Normal from double exponential

For the double exponential density $g(x)=e^{-|x|}/2$ and standard normal density

$$f(x)=\frac{1}{\sqrt{2\pi}}\exp(-x^2/2),$$

the ratio satisfies

$$\frac{f(x)}{g(x)}=\sqrt{\frac2\pi}e^{-x^2/2+|x|}\le\sqrt{\frac{2e}{\pi}}\approx1.3155\equiv c.$$

The rejection test may be written as

$$u>\exp\left(-\tfrac12x^2+|x|-\tfrac12\right)=\exp\left(-\tfrac12(|x|-1)^2\right).$$

Using symmetry: generate U_1,U_2,U_3, set $X=-\log U_1$, reject if $U_2>\exp[-0.5(X-1)^2]$, then set $X\leftarrow -X$ if $U_3\le0.5$.

2.5 Normal Random Variables and Vectors

Multivariate normal

A d-dimensional normal is written $N(\mu,\Sigma)$, where μ is a d-vector and Σ is a $d\times d$ covariance matrix. A covariance matrix must be symmetric and positive semidefinite:

$$x^\top\Sigma x\ge0, \qquad x\in\mathbb R^d.$$

If Σ is positive definite, the density is

$$\phi_{\mu,\Sigma}(x)=\frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}}\exp\left[-\tfrac12(x-\mu)^\top\Sigma^{-1}(x-\mu)\right].$$

If $X\sim N(\mu,\Sigma)$, then $X_i\sim N(\mu_i,\sigma_i^2)$ with $\sigma_i^2=\Sigma_{ii}$, and

$$\mathrm{Cov}(X_i,X_j)=\mathbb E[(X_i-\mu_i)(X_j-\mu_j)]=\Sigma_{ij},$$

$$\rho_{ij}=\frac{\Sigma_{ij}}{\sigma_i\sigma_j}, \text{ equivalently } \Sigma_{ij}=\sigma_i\sigma_j\rho_{ij}.$$

If Σ is positive semidefinite but singular, define $X=\mu+AZ$ with $Z\sim N(0,I_d)$ and $AA^\top=\Sigma$. Then X has no full d-dimensional density when $\mathrm{rank}(\Sigma)<d$.

Useful properties

Linear transformation: if $X\sim N(\mu,\Sigma)$, then

$$AX\sim N(A\mu,A\Sigma A^\top).$$

Conditioning formula: for a partitioned normal vector with $\Sigma_{[22]}$ full rank,

$$(X_{[1]}\mid X_{[2]}=x)\sim N\left(\mu_{[1]}+\Sigma_{[12]}\Sigma_{[22]}^{-1}(x-\mu_{[2]}),\ \Sigma_{[11]}-\Sigma_{[12]}\Sigma_{[22]}^{-1}\Sigma_{[21]}\right).$$

Moment generating function:

$$\mathbb E[\exp(\theta^\top X)]=\exp(\mu^\top\theta+\tfrac12\theta^\top\Sigma\theta).$$

2.6 Generating Normals

Box-Muller method

The Box-Muller method generates a bivariate standard normal. If $Z\sim N(0,I_2)$, then $R=Z_1^2+Z_2^2$ is exponential with mean 2:

$$P(R\le x)=1-e^{-x/2}.$$

Given R, (Z_1,Z_2) is uniformly distributed on the circle of radius $\sqrt R$. Generate independent U_1,U_2, set

$$R=-2\log U_1, \qquad V=2\pi U_2,$$

and return

$$Z_1=\sqrt R\cos V, \qquad Z_2=\sqrt R\sin V.$$

Multivariate normals

To sample $X\sim N(\mu,\Sigma)$, generate $Z\sim N(0,I_d)$, find A such that

$$AA^\top=\Sigma,$$

and set $X=\mu+AZ$. Then $X\sim N(\mu,\Sigma)$.

2.7 Cholesky Factorization

Positive definite case

Cholesky represents $\Sigma=AA^\top$ with A lower triangular. For

$$\Sigma=\begin{pmatrix}\sigma_1^2&\sigma_1\sigma_2\rho\\ \sigma_1\sigma_2\rho&\sigma_2^2\end{pmatrix},$$

a Cholesky factor is

$$A=\begin{pmatrix}\sigma_1&0\\ \rho\sigma_2&\sqrt{1-\rho^2}\sigma_2\end{pmatrix}.$$

Thus

$$X_1=\mu_1+\sigma_1Z_1,$$

$$X_2=\mu_2+\sigma_2\rho Z_1+\sigma_2\sqrt{1-\rho^2}Z_2.$$

For general d, use

$$\Sigma_{ij}=\sum_{k=1}^j A_{ik}A_{jk}, \qquad j\le i.$$

The recursive formulas are

$$A_{ij}=\frac{\Sigma_{ij}-\sum_{k=1}^{j-1}A_{ik}A_{jk}}{A_{jj}}, \qquad j<i,$$

$$A_{ii}=\sqrt{\Sigma_{ii}-\sum_{k=1}^{i-1}A_{ik}^2}.$$

Loop over columns $j=1,\dots,d$ and rows $i=j,\dots,d$, solving one new entry at a time.

Semidefinite case

If Σ is only positive semidefinite, $\mathrm{rank}(\Sigma)<d$ and some diagonal entry of a lower triangular factor must be zero, so Cholesky may divide by zero or become round-off sensitive. A mathematical fix sets the whole jth column of A to zero when $A_{jj}=0$, but exact zero checks are fragile.

Prefer reducing to full rank: choose a subvector $\tilde X=(X_{i_1},\dots,X_{i_k})$ with full-rank covariance $\tilde\Sigma$, find $\tilde A\tilde A^\top=\tilde\Sigma$, and sample

$$X=D\tilde AZ, \qquad Z\sim N(0,I_k).$$

3 Generating Sample Paths

3.1 Brownian Motion

One-dimensional BM

A standard Brownian motion $W(t)$ on $[0,T]$ satisfies $W(0)=0$, continuous sample paths, independent increments, and

$$W(t)-W(s)\sim N(0,t-s), \qquad 0\le s<t\le T.$$

Hence $W(t)\sim N(0,t)$. A Brownian motion with drift μ and diffusion coefficient σ^2, written $X\sim\mathrm{BM}(\mu,\sigma^2)$, can be represented by

$$X(t)=X(0)+\mu t+\sigma W(t),$$

so $X(t)\sim N(X(0)+\mu t,\sigma^2 t)$ and

$$dX(t)=\mu\,dt+\sigma\,dW(t).$$

With deterministic time-varying coefficients,

$$dX(t)=\mu(t)\,dt+\sigma(t)\,dW(t),$$

$$X(t)=X(0)+\int_0^t\mu(s)\,ds+\int_0^t\sigma(s)\,dW(s).$$

For $0\le s<t$,

$$\mathbb E[X(t)-X(s)]=\int_s^t\mu(u)\,du,$$

$$\mathrm{var}[X(t)-X(s)]=\int_s^t\sigma^2(u)\,du.$$

Random-walk construction

For dates $0=t_0<t_1<\cdots<t_n$, let $Z_1,\dots,Z_n$ be independent $N(0,1)$. For standard BM,

$$W(t_{i+1})=W(t_i)+\sqrt{t_{i+1}-t_i}\,Z_{i+1}, \qquad i=0,\dots,n-1.$$

For $X\sim\mathrm{BM}(\mu,\sigma^2)$ with constant coefficients,

$$X(t_{i+1})=X(t_i)+\mu(t_{i+1}-t_i)+\sigma\sqrt{t_{i+1}-t_i}\,Z_{i+1}.$$

For time-dependent coefficients, the exact grid recursion is

$$X(t_{i+1})=X(t_i)+\int_{t_i}^{t_{i+1}}\mu(s)\,ds+\sqrt{\int_{t_i}^{t_{i+1}}\sigma^2(u)\,du}\,Z_{i+1}.$$

These recursions are exact at the grid points. Discretization error appears if the path is deterministically interpolated between grid points. The Euler approximation

$$X(t_{i+1})=X(t_i)+\mu(t_i)(t_{i+1}-t_i)+\sigma(t_i)\sqrt{t_{i+1}-t_i}\,Z_{i+1}$$

is generally biased even at the grid dates when μ,σ vary in time, since it uses approximate increment mean and variance.

Multivariate normal view

The vector $(W(t_1),\dots,W(t_n))$ is multivariate normal with mean 0 and covariance matrix C satisfying

$$C_{ij}=\mathrm{Cov}(W(t_i),W(t_j))=\min(t_i,t_j).$$

Thus one may simulate it as AZ, where $Z\sim N(0,I_n)$ and $AA^\top=C$. For the Brownian covariance matrix, a lower triangular factor is compactly

$$A_{ij}=\sqrt{t_j-t_{j-1}}\,\mathbf 1_{\{i\ge j\}}, \qquad t_0=0.$$

This Cholesky method is the matrix form of the random-walk construction. Direct multiplication is $O(n^2)$, while the recursive construction is $O(n)$.

For $X\sim\mathrm{BM}(\mu,\sigma^2)$, the mean vector has entries μt_i, the covariance matrix is σ^2C, and the factor is σA.

3.2 Multidimensional Brownian Motion

Definition and representation

A standard d-dimensional Brownian motion $W(t)=(W_1(t),\dots,W_d(t))^\top$ has

$$W(t)-W(s)\sim N(0,(t-s)I).$$

Thus each coordinate is standard BM and different coordinates are independent.

For $\mu\in\mathbb R^d$ and positive semidefinite Σ, $X\sim\mathrm{BM}(\mu,\Sigma)$ means

$$X(t)-X(s)\sim N((t-s)\mu,(t-s)\Sigma).$$

If $BB^\top=\Sigma$ and W is standard BM, then

$$X(t)=X(0)+\mu t+BW(t),$$

and $dX(t)=\mu\,dt+B\,dW(t)$. The law depends on B only through $BB^\top$.

Time-varying coefficients

For

$$dX(t)=\mu(t)\,dt+B(t)\,dW(t), \qquad B(t)B(t)^\top=\Sigma(t),$$

the increment distribution is

$$X(t)-X(s)\sim N\left(\int_s^t\mu(u)\,du,\int_s^t\Sigma(u)\,du\right).$$

For $X\sim\mathrm{BM}(\mu,\Sigma)$,

$$\mathrm{Cov}[X_i(s),X_j(t)]=\min(s,t)\Sigma_{ij}.$$

Simulation

Let $Z_1,Z_2,\dots$ be independent $N(0,I)$ vectors. Standard d-dimensional BM is simulated by

$$W(t_{i+1})=W(t_i)+\sqrt{t_{i+1}-t_i}\,Z_{i+1}.$$

For $X\sim\mathrm{BM}(\mu,\Sigma)$, factor $\Sigma=BB^\top$ and use

$$X(t_{i+1})=X(t_i)+\mu(t_{i+1}-t_i)+\sqrt{t_{i+1}-t_i}\,BZ_{i+1}.$$

For time-dependent covariance, use

$$X(t_{i+1})=X(t_i)+\int_{t_i}^{t_{i+1}}\mu(s)\,ds+B_iZ_{i+1},$$

where

$$B_iB_i^\top=\int_{t_i}^{t_{i+1}}\Sigma(u)\,du.$$

This may require a separate factorization for each time step.

3.3 Geometric Brownian Motion

Exact solution and recursion

A geometric Brownian motion is exponential Brownian motion:

$$\frac{dS(t)}{S(t)}=\mu\,dt+\sigma\,dW(t), \qquad S\sim\mathrm{GBM}(\mu,\sigma^2).$$

The exact solution is

$$S(t)=S(0)\exp\left(\left[\mu-\tfrac12\sigma^2\right]t+\sigma W(t)\right).$$

For $u<t$,

$$S(t)=S(u)\exp\left(\left[\mu-\tfrac12\sigma^2\right](t-u)+\sigma(W(t)-W(u))\right).$$

Hence for $0=t_0<t_1<\cdots<t_n$,

$$S(t_{i+1})=S(t_i)\exp\left(\left[\mu-\tfrac12\sigma^2\right]\Delta t_i+\sigma\sqrt{\Delta t_i}\,Z_{i+1}\right),$$

where $\Delta t_i=t_{i+1}-t_i$. This is exact at the simulated dates and has no model discretization error.

Lognormal distribution

Write $Y\sim LN(\mu,\sigma^2)$ if $Y=\exp(\mu+\sigma Z)$ with $Z\sim N(0,1)$. Then

$$P(Y\le y)=\Phi\left(\frac{\log y-\mu}{\sigma}\right),$$

$$f(y)=\frac{1}{y\sigma\sqrt{2\pi}}\exp\left(-\frac{(\log y-\mu)^2}{2\sigma^2}\right).$$

Using $\mathbb E[e^{aZ}]=e^{a^2/2}$,

$$\mathbb E[Y]=e^{\mu+\sigma^2/2}, \qquad \mathrm{var}(Y)=e^{2\mu+\sigma^2}(e^{\sigma^2}-1).$$

Also $P(Y\le e^\mu)=1/2$, so e^μ is the median. For $S\sim\mathrm{GBM}(\mu,\sigma^2)$,

$$\frac{S(t)}{S(0)}\sim LN\left(\left[\mu-\tfrac12\sigma^2\right]t,\sigma^2t\right),$$

$$\mathbb E[S(t)]=e^{\mu t}S(0), \qquad \mathrm{var}[S(t)]=e^{2\mu t}S^2(0)(e^{\sigma^2t}-1).$$

The Markov conditional mean is

$$\mathbb E[S(t)\mid S(\tau),0\le\tau\le u]=\mathbb E[S(t)\mid S(u)]=e^{\mu(t-u)}S(u).$$

3.4 Risk-Neutral Dynamics

Assets and futures

With constant continuously compounded rate r, $\beta(t)=e^{rt}$. Under the risk-neutral measure, for a non-dividend-paying asset,

$$\frac{S(u)}{\beta(u)}=\mathbb E\left[\frac{S(t)}{\beta(t)}\mid\{S(\tau),0\le\tau\le u\}\right].$$

Thus the risk-neutral GBM dynamics are

$$\frac{dS(t)}{S(t)}=r\,dt+\sigma\,dW(t).$$

With continuous dividend yield δ,

$$\frac{dS(t)}{S(t)}=(r-\delta)\,dt+\sigma\,dW(t).$$

For a futures price $F(t,T)$ with fixed maturity T, the risk-neutral futures price is a martingale. If modeled by GBM,

$$\frac{dF(t,T)}{F(t,T)}=\sigma\,dW(t).$$

3.5 Path-Dependent Payoffs

Asian options

For monitoring dates $t_1,\dots,t_n$, the arithmetic average is

$$\bar S=\frac1n\sum_{i=1}^n S(t_i).$$

Asian call and put payoffs are $(\bar S-K)^+$ and $(K-\bar S)^+$. Floating-strike variants include $(S(T)-\bar S)^+$ and $(\bar S-S(T))^+$. The arithmetic average distribution is generally intractable, so simulation is natural.

For continuous monitoring over $[u,t]$,

$$\bar S=\frac{1}{t-u}\int_u^t S(\tau)\,d\tau.$$

Geometric average options

The discrete geometric average is

$$G=\left(\prod_{i=1}^n S(t_i)\right)^{1/n}.$$

Because lognormal variables remain lognormal under geometric averaging, these options are analytically convenient. Under risk-neutral GBM with dividend yield δ,

$$G=S(0)\exp\left(\left[r-\delta-\tfrac12\sigma^2\right]\frac1n\sum_{i=1}^n t_i+\frac\sigma n\sum_{i=1}^n W(t_i)\right).$$

Since $\mathrm{Cov}(W(s),W(t))=\min(s,t)$,

$$\sum_{i=1}^n W(t_i)\sim N\left(0,\sum_{i=1}^n(2i-1)t_{n+1-i}\right).$$

The geometric average has the same distribution as a GBM value at

$$T_G=\frac1n\sum_{i=1}^n t_i,$$

with adjusted volatility and dividend yield

$$\bar\sigma^2=\frac{\sigma^2}{n^2T_G}\sum_{i=1}^n(2i-1)t_{n+1-i},$$

$$\bar\delta=\delta+\tfrac12\sigma^2-\tfrac12\bar\sigma^2.$$

Thus a geometric-average option can be priced using Black-Scholes with adjusted parameters. The continuous geometric average

$$\exp\left(\frac{1}{t-u}\int_u^t\log S(\tau)\,d\tau\right)$$

is also lognormally distributed.

Barrier and lookback options

A discretely monitored down-and-out call with barrier b has payoff

$$\mathbf 1_{\{\tau(b)>T\}}(S(T)-K)^+, \qquad \tau(b)=\inf\{t_i:S(t_i)<b\}.$$

If no grid value crosses below b, set $\tau(b)=\infty$. A down-and-in call has payoff

$$\mathbf 1_{\{\tau(b)\le T\}}(S(T)-K)^+.$$

For continuous monitoring, use $\tilde\tau(b)=\inf\{t\ge0:S(t)\le b\}$. Discretely monitored barrier options are priced by simulating $S(t_1),\dots,S(t_n),S(T)$.

Lookback options depend on path extrema. Discrete lookback put and call payoffs at t_n are

$$\max_{i=1,\dots,n}S(t_i)-S(t_n), \qquad S(t_n)-\min_{i=1,\dots,n}S(t_i).$$

3.6 Multi-Asset GBM

Correlated assets

Let X_i be standard Brownian motions with correlations ρ_{ij}. A multidimensional GBM is

$\frac{dS_i(t)}{S_i(t)} = \mu_i\,dt + \sigma_i\,dX_i(t), \qquad i = 1,\dots,d.$

Define $\Sigma_{ij} = \sigma_i\sigma_j\rho_{ij}$. Then $(\sigma_1 X_1,\dots,\sigma_d X_d) \sim \mathrm{BM}(0,\Sigma)$ and $S = (S_1,\dots,S_d) \sim \mathrm{GBM}(\mu,\Sigma)$.

The price covariance is

$\mathrm{Cov}[S_i(t),S_j(t)] = S_i(0)S_j(0)e^{(\mu_i+\mu_j)t}\left(e^{\rho_{ij}\sigma_i\sigma_j t} - 1\right).$

If $AA^\top = \Sigma$ and a_i is the ith row of A,

$\frac{dS_i(t)}{S_i(t)} = \mu_i\,dt + a_i\,dW(t) = \mu_i\,dt + \sum_{j=1}^d A_{ij}\,dW_j(t).$

Simulation and payoffs

With independent $Z_k = (Z_{k1},\dots,Z_{kd}) \sim N(0,I)$,

$S_i(t_{k+1}) = S_i(t_k)\exp\left(\left[\mu_i - \frac{1}{2}\sigma_i^2\right]\Delta t_k + \sqrt{\Delta t_k}\sum_{j=1}^d A_{ij}Z_{k+1,j}\right).$

Under the risk-neutral measure, $\mu_i = r - \delta_i$. Typical multi-asset payoffs:

Spread option:

$([S_1(T) - S_2(T)] - K)^+.$

Basket option:

$([c_1S_1(T) + c_2S_2(T) + \cdots + c_dS_d(T)] - K)^+.$

Outperformance option:

$(\max\{c_1S_1(T), c_2S_2(T),\dots,c_dS_d(T)\} - K)^+.$

Two-asset barrier example:

$\mathbf{1}_{\{\min_{i=1,\dots,n} S_2(t_i) < b\}}(K - S_1(T))^+.$

4 Variance Reduction Techniques

4.1 Control Variates

Basic method

Let $Y_1,\dots,Y_n$ be i.i.d. outputs for estimating $\mathbb{E}[Y]$, and suppose X_i is generated with Y_i and $\mathbb{E}[X]$ is known. For fixed b define

$Y_i(b) = Y_i - b(X_i - \mathbb{E}[X]),$

and estimate

$\overline{Y}(b) = \overline{Y} - b(\overline{X} - \mathbb{E}[X]) = \frac{1}{n}\sum_{i=1}^n\{Y_i - b(X_i - \mathbb{E}[X])\}.$

This is unbiased for all b. Its per-replication variance is

$\sigma^2(b) = \mathrm{var}[Y_i(b)] = \sigma_Y^2 - 2b\sigma_X\sigma_Y\rho_{XY} + b^2\sigma_X^2.$

The optimal coefficient is

$b^* = \frac{\sigma_Y}{\sigma_X}\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{var}(X)}.$

The resulting variance ratio relative to plain Monte Carlo is

$\frac{\mathrm{var}[\overline{Y} - b^*(\overline{X} - \mathbb{E}[X])]}{\mathrm{var}[\overline{Y}]} = 1 - \rho_{XY}^2.$

Thus speed-up is approximately $1/(1-\rho_{XY}^2)$. The sign of ρ_{XY} is absorbed by b^*; only $|\rho_{XY}|$ matters.

Estimated coefficient and CI

In practice use the sample regression coefficient

$\hat{b}_n = \frac{\sum_{i=1}^n(X_i - \overline{X})(Y_i - \overline{Y})}{\sum_{i=1}^n(X_i - \overline{X})^2}.$

Then $\hat{b}_n \to b^*$ a.s. and

$\overline{Y}(\hat{b}_n) = \frac{1}{n}\sum_{i=1}^n\{Y_i - \hat{b}_n(X_i - \mathbb{E}[X])\}.$

Using $\hat{b}_n$ creates finite-sample bias, but has the same asymptotic precision as using b^*. For fixed b,

$\frac{\overline{Y}(b) - \mathbb{E}[Y]}{\sigma(b)/\sqrt{n}} \Rightarrow N(0,1),$

with sample standard deviation

$s(b) = \left[\frac{1}{n-1}\sum_{i=1}^n(Y_i(b) - \overline{Y}(b))^2\right]^{1/2}.$

Hence an asymptotic CI is $\overline{Y}(b) \pm z_{\delta/2}s(b)/\sqrt{n}$. Also

$\frac{\overline{Y}(\hat{b}_n) - \mathbb{E}[Y]}{s(\hat{b}_n)/\sqrt{n}} \Rightarrow N(0,1).$

Financial controls

Any martingale with known initial value is a natural control. Under risk-neutral pricing with constant r, $e^{-rt}S(t)$ is a martingale, so $\mathbb{E}[e^{-rT}S(T)] = S(0)$. For discounted payoff Y,

$\frac{1}{n}\sum_{i=1}^n\{Y_i - b[e^{-rT}S_i(T) - S(0)]\}$

uses the stock as control. For a call, this works best near $K = 0$ and may weaken for deep OTM strikes. A stronger control can be a tractable option. For Asian options,

$\overline{S}_A = \frac{1}{n}\sum_{i=1}^n S(t_i), \qquad \overline{S}_G = \left(\prod_{i=1}^n S(t_i)\right)^{1/n}.$

Under GBM, $\overline{S}_G$ is lognormal and the geometric-average Asian option can be priced analytically; its payoff is often highly correlated with the arithmetic-average Asian payoff.

4.2 Antithetic Variates

Estimator

Antithetic variates pair inputs to create negative dependence while preserving the marginal law. If $U \sim \mathrm{Unif}[0,1]$, use U and $1-U$; for a symmetric normal input, use Z and $-Z$.

Let $(Y_i,\tilde{Y}_i)$ be i.i.d. pairs with the same marginal distribution. The estimator is

$\hat{Y}_{AV} = \frac{1}{2n}\left(\sum_{i=1}^n Y_i + \sum_{i=1}^n \tilde{Y}_i\right) = \frac{1}{n}\sum_{i=1}^n\frac{Y_i + \tilde{Y}_i}{2}.$

Apply the CLT to pair averages:

$\frac{\hat{Y}_{AV} - \mathbb{E}[Y]}{\sigma_{AV}/\sqrt{n}} \Rightarrow N(0,1), \qquad \sigma_{AV}^2 = \mathrm{var}\left(\frac{Y_i + \tilde{Y}_i}{2}\right).$

A CI is $\hat{Y}_{AV} \pm z_{\delta/2}s_{AV}/\sqrt{n}$, where s_{AV} is the sample sd of pair averages.

When it helps

Compare n antithetic pairs with $2n$ independent samples. Since

$\mathrm{var}[Y_i + \tilde{Y}_i] = 2\,\mathrm{var}[Y_i] + 2\,\mathrm{Cov}(Y_i,\tilde{Y}_i),$

antithetic sampling reduces variance iff $\mathrm{Cov}(Y_i,\tilde{Y}_i) < 0$. Monotone outputs often satisfy this when paired by $U, 1-U$ or $Z, -Z$.

GBM call example

For a call, pair Z_i and $-Z_i$:

$S_i^+ = S_0\exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\right),$

$S_i^- = S_0\exp\left((r - \frac{1}{2}\sigma^2)T - \sigma\sqrt{T}Z_i\right).$

Set $X_i = e^{-rT}(S_i^+ - K)^+$ and $Y_i = e^{-rT}(S_i^- - K)^+$. Then

$\hat{v} = \frac{1}{n}\sum_{i=1}^n\frac{X_i + Y_i}{2},$

$\mathrm{SE} = \left[\frac{1}{n(n-1)}\left\{\sum_{i=1}^n\left(\frac{X_i + Y_i}{2}\right)^2 - n\hat{v}^2\right\}\right]^{1/2}.$

4.3 Stratified Sampling

Basic method

Let $A_1,\dots,A_K$ be disjoint strata with probabilities p_i. Then

$\mathbb{E}[Y] = \sum_{i=1}^K p_i\mathbb{E}[Y \mid Y \in A_i].$

With proportional allocation $n_i = np_i$ and conditional draws $Y_{ij} \sim (Y \mid Y \in A_i)$,

$\hat{Y} = \sum_{i=1}^K p_i\frac{1}{n_i}\sum_{j=1}^{n_i}Y_{ij} = \frac{1}{n}\sum_{i=1}^K\sum_{j=1}^{n_i}Y_{ij}.$

Stratification removes the random variation in the number of observations landing in each stratum.

General allocation

Often strata are defined by another variable X. If A_i partition the support of X,

$\mathbb{E}[Y] = \sum_{i=1}^K p_i\mathbb{E}[Y \mid X \in A_i].$

For $n_i \geq 1$, $\sum_i n_i = n$, and $q_i = n_i/n$,

$\hat{Y} = \sum_{i=1}^K p_i\frac{1}{n_i}\sum_{j=1}^{n_i}Y_{ij} = \frac{1}{n}\sum_{i=1}^K\frac{p_i}{q_i}\sum_{j=1}^{n_i}Y_{ij}.$

Let $\mu_i = \mathbb{E}[Y \mid X \in A_i]$ and $\sigma_i^2 = \mathrm{var}(Y \mid X \in A_i)$. Then $\hat{Y}$ is unbiased and

$\mathrm{var}[\hat{Y}] = \sum_{i=1}^K p_i^2\frac{\sigma_i^2}{n_i} = \frac{\sigma^2(q)}{n}, \qquad \sigma^2(q) = \sum_{i=1}^K\frac{p_i^2}{q_i}\sigma_i^2.$

With fixed K, $\sqrt{n}(\hat{Y} - \mu) \Rightarrow N(0,\sigma^2(q))$, and

$s^2(q) = \sum_{i=1}^K\frac{p_i^2}{q_i}s_i^2$

estimates $\sigma^2(q)$.

Uniform and inverse-transform strata

For $U \sim \mathrm{Unif}[0,1]$, equal strata are $A_i = ((i-1)/n, i/n)$. With $U_i \sim \mathrm{Unif}[0,1]$,

$V_i = \frac{i-1}{n} + \frac{U_i}{n}, \qquad \hat{Y} = \frac{1}{n}\sum_{i=1}^n f(V_i).$

For a distribution with CDF F, choose probabilities p_i and quantiles $a_i = F^{-1}(p_1 + \cdots + p_i)$. Conditional simulation in stratum $(a_{i-1}, a_i]$ uses

$V = F(a_{i-1}) + U\{F(a_i) - F(a_{i-1})\}, \qquad F^{-1}(V) \sim (Y \mid Y \in A_i).$

Proportional and optimal allocation

For proportional allocation $q_i = p_i$,

$\sigma^2(q) = \sum_{i=1}^K p_i\sigma_i^2.$

Plain variance decomposes as

$\mathrm{var}(Y) = \sum_{i=1}^K p_i\sigma_i^2 + \sum_{i=1}^K p_i\mu_i^2 - \left(\sum_{i=1}^K p_i\mu_i\right)^2.$

By Jensen, proportional stratification cannot increase variance, with strict gain unless all μ_i are equal. Optimizing $\sigma^2(q)$ gives

$q_i^* = \frac{p_i\sigma_i}{\sum_{\ell=1}^K p_\ell\sigma_\ell}, \qquad \sigma^2(q^*) = \left(\sum_{i=1}^K p_i\sigma_i\right)^2.$

Thus allocate samples proportional to stratum probability times within-stratum sd.

Call option strata

For a GBM call, write $W_T = \sqrt{T}\Phi^{-1}(U)$ with $U \sim \mathrm{Unif}[0,1]$ and

$h(u) = \left(S_0\exp\left\{-\frac{1}{2}\sigma^2T + \sigma\sqrt{T}\Phi^{-1}(u)\right\} - e^{-rT}K\right)^+.$

With k equal strata, $U_{ij} = (i-1)/k + V_{ij}/k$, $\hat{\mu}_i = n_i^{-1}\sum_{j=1}^{n_i}h(U_{ij})$, and

$\hat{\mu} = \frac{1}{k}\sum_{i=1}^k\hat{\mu}_i, \qquad \mathrm{SE} = \frac{1}{k}\left(\sum_{i=1}^k\frac{s_i^2}{n_i}\right)^{1/2}.$

4.4 Importance Sampling

Change of measure

To estimate $\mu = \mathbb{E}[h(X)]$ with density f, choose an alternative density g such that $g(x) = 0$ implies $f(x) = 0$. If $Y \sim g$,

$\mu = \int h(x)f(x)\,dx = \int h(x)\frac{f(x)}{g(x)}g(x)\,dx = \mathbb{E}_g\left[h(Y)\frac{f(Y)}{g(Y)}\right].$

The likelihood ratio is $f(Y)/g(Y)$, and the estimator averages $h(Y_i)f(Y_i)/g(Y_i)$. In a discrete model, use $h(Y_i)p(Y_i)/\overline{p}(Y_i)$.

Variance and zero-variance density

The IS estimator has variance

$\frac{1}{n}\mathrm{var}_g\left[h(Y)\frac{f(Y)}{g(Y)}\right].$

For $h \geq 0$, the zero-variance density is

$g^*(x) = c\,h(x)f(x), \qquad \frac{1}{c} = \int h(x)f(x)\,dx = \mu.$

It is unusable directly because it requires μ, but it says g should mimic $h(x)f(x)$.

Normal mean shift and options

For $X \sim N(0,1)$, choose $Y \sim N(\theta,1)$. The likelihood ratio is

$\frac{f(x)}{g(x)} = \exp\left(-\theta x + \frac{1}{2}\theta^2\right).$

Mode matching sets $\theta = \arg\max_x h(x)f(x)$. For a binary call,

$\mathbb{E}[e^{-rT}\mathbf{1}_{\{S_T \geq K\}}] = \mathbb{E}[e^{-rT}\mathbf{1}_{\{X \geq b\}}],$

where

$b = \frac{\log(K/S_0) - (r - \sigma^2/2)T}{\sigma\sqrt{T}}.$

Here $h(x) = e^{-rT}\mathbf{1}_{\{x \geq b\}}$, $x^* = \max\{b, 0\}$, and with $Y \sim N(x^*, 1)$ the output is

$h(Y)\frac{f(Y)}{g(Y)} = \begin{cases} 0, & Y < b, \\ \exp\{-rT - x^*Y + (x^*)^2/2\}, & Y \geq b. \end{cases}$

For a vanilla call,

$h(x) = \left(S_0\exp\left\{-\frac{1}{2}\sigma^2T + \sigma\sqrt{T}x\right\} - e^{-rT}K\right)^+,$

and the shift pushes sampling toward exercise states.

Exponential tilting

For density f, the exponential tilt family is

$f_\theta(x) = \frac{e^{\theta x}f(x)}{\mathbb{E}[e^{\theta X}]}, \qquad \theta \in \mathbb{R}.$

For a pmf p,

$p_\theta(x) = \frac{e^{\theta x}p(x)}{\mathbb{E}[e^{\theta X}]}.$

For normal X, this is mean shifting. For rare large losses, take $\theta > 0$.

Bernoulli rare event

Let independent defaults satisfy $P(X_k = 1) = p_k$ and $L = \sum_{k=1}^m X_k$. To estimate $P(L > x)$ with $x > \mathbb{E}[L]$, tilt each Bernoulli:

$\overline{p}_k = P(Y_k = 1) = \frac{p_ke^\theta}{1 + p_k(e^\theta - 1)}, \qquad \overline{L} = \sum_{k=1}^m Y_k.$

The IS output is

$H = \mathbf{1}_{\{\overline{L} > x\}}\prod_{k=1}^m\left(\frac{p_k}{\overline{p}_k}\right)^{Y_k}\left(\frac{1-p_k}{1-\overline{p}_k}\right)^{1-Y_k}.$

With

$\phi(\theta) = \sum_{k=1}^m\log[1 + p_k(e^\theta - 1)],$

one has $E[H^2] \leq e^{-2\theta x + 2\phi(\theta)}$. Choose $\theta > 0$ by solving $\phi'(\theta) = x$.

4.5 Cross-Entropy Method

Principle

Cross-entropy selects a parametric IS density f_θ close to the zero-variance density

$g^*(x) = \frac{1}{\mu}h(x)f(x).$

Minimize

$R(g^* \| f_\theta) = \int\log\left(\frac{g^*(x)}{f_\theta(x)}\right)g^*(x)\,dx,$

equivalently maximize

$\int h(x)f(x)\log f_\theta(x)\,dx.$

The population first-order equation is

$\mathbb{E}\left[h(X)\frac{\partial}{\partial\theta}\log f_\theta(X)\right] = 0.$

Basic algorithm

Using pilot samples $X_1,\dots,X_N \sim f$, solve

$\frac{1}{N}\sum_{k=1}^N h(X_k)\frac{\partial}{\partial\theta}\log f_\theta(X_k) = 0.$

For $f = N(0, I_m)$ and $f_\theta = N(\theta, I_m)$,

$\hat{\theta} = \frac{\sum_{k=1}^N h(X_k)X_k}{\sum_{k=1}^N h(X_k)}.$

Then sample $Y_i \sim f_{\hat{\theta}}$ and average

$H_i = h(Y_i)\frac{f(Y_i)}{f_{\hat{\theta}}(Y_i)}.$

For a one-dimensional normal shift, $H_i = h(Y_i)\exp\{-\hat{\theta}Y_i + \hat{\theta}^2/2\}$. For very rare events, the basic method may fail because $\sum_k h(X_k) = 0$ in the pilot run.

Iterative cross-entropy

Let $\ell_\theta(x) = f(x)/f_\theta(x)$. If pilot samples are drawn from f_v, then maximize

$\mathbb{E}_v[h(Y)\ell_v(Y)\log f_\theta(Y)],$

and solve

$\mathbb{E}_v\left[h(Y)\ell_v(Y)\frac{\partial}{\partial\theta}\log f_\theta(Y)\right] = 0.$

At iteration j, sample $Y_k \sim f_{\hat{\theta}^j}$ and update from

$\frac{1}{N}\sum_{k=1}^N h(Y_k)\ell_{\hat{\theta}^j}(Y_k)\frac{\partial}{\partial\theta}\log f_\theta(Y_k) = 0.$

For $N(\theta, I_m)$,

$\hat{\theta}^{j+1} = \frac{\sum_{k=1}^N h(Y_k)\ell_{\hat{\theta}^j}(Y_k)Y_k}{\sum_{k=1}^N h(Y_k)\ell_{\hat{\theta}^j}(Y_k)}.$

Here

$\ell_{\hat{\theta}^j}(Y_k) = \exp(-\langle\hat{\theta}^j, Y_k\rangle + \|\hat{\theta}^j\|^2/2).$

Usually only a few iterations are needed; convergence means the updates oscillate in a small range rather than become deterministic.

5 Discretization Methods

5.1 Euler And Milstein Schemes

Euler scheme

For the SDE $dX(t) = a(X(t))\,dt + b(X(t))\,dW(t)$, with $X \in \mathbb{R}^d$ and $W \in \mathbb{R}^m$, the Euler approximation on $t_i = ih$ is

$\hat{X}(i+1) = \hat{X}(i) + a(\hat{X}(i))h + b(\hat{X}(i))\sqrt{h}Z_{i+1},$

where Z_i are independent m-dimensional standard normals. Euler freezes the drift and diffusion integrands on each interval. The drift is order h, while the diffusion increment is order $\sqrt{h}$, so a first refinement focuses on the diffusion term.

Scalar Milstein refinement

In the scalar case, Ito's formula gives

$db(X(t)) = \left[b'(X(t))a(X(t)) + \frac{1}{2}b''(X(t))b^2(X(t))\right]dt$

$+ b'(X(t))b(X(t))\,dW(t).$

Dropping the higher-order drift part,

$b(X(u)) \approx b(X(t)) + b'(X(t))b(X(t))[W(u) - W(t)].$

With $\Delta W = W(t+h) - W(t)$,

$\int_t^{t+h} b(X(u))\,dW(u) \approx b(X(t))\Delta W +$

$b'(X(t))b(X(t))\int_t^{t+h}[W(u) - W(t)]\,dW(u).$

The remaining integral is

$\int_t^{t+h}[W(u) - W(t)]\,dW(u) = \frac{1}{2}(\Delta W)^2 - \frac{1}{2}h.$

Thus the Milstein one-step approximation is

$X(t+h) \approx X(t) + a(X(t))h + b(X(t))\Delta W + \frac{1}{2}b'(X(t))b(X(t))\{(\Delta W)^2 - h\}.$

Equivalently, with $\Delta W = \sqrt{h}Z_{i+1}$,

$\hat{X}(i+1) = \hat{X}(i) + a(\hat{X}(i))h + b(\hat{X}(i))\sqrt{h}Z_{i+1} + \frac{1}{2}b'(\hat{X}(i))b(\hat{X}(i))h(Z_{i+1}^2 - 1).$

Conditional on $\hat{X}(i)$, the extra term has mean 0 and is uncorrelated with the Euler diffusion term since $Z^2 - 1$ and Z are uncorrelated.

GBM comparison

For $dX_t = rX_t\,dt + \sigma X_t\,dW_t$, Euler is

$\hat{X}_{t_{i+1}} = \hat{X}_{t_i} + r\hat{X}_{t_i}h + \sigma\hat{X}_{t_i}\sqrt{h}Z_{i+1}.$

Milstein is

$\hat{X}_{t_{i+1}} = \hat{X}_{t_i} + r\hat{X}_{t_i}h + \sigma\hat{X}_{t_i}\sqrt{h}Z_{i+1} + \frac{1}{2}\sigma^2\hat{X}_{t_i}h(Z_{i+1}^2 - 1).$

The exact benchmark is

$X_{t_{i+1}} = X_{t_i}\exp\{(r - \frac{1}{2}\sigma^2)h + \sigma\sqrt{h}Z_{i+1}\}.$

In the lecture's numerical comparison, Euler strong error is order $\sqrt{h}$, while Milstein error is order h.

Multidimensional difficulty

For components X_i, W_j, a_i, and b_{ij},

$X_i(t+h) = X_i(t) + \int_t^{t+h}a_i(X(u))\,du + \sum_{j=1}^m\int_t^{t+h}b_{ij}(X(u))\,dW_j(u).$

The diffusion refinement contains

$\int_t^{t+h}b_{ij}(X(u))\,dW_j(u) \approx b_{ij}(X(t))\Delta W_j$

$+ \sum_{\ell=1}^d\sum_{k=1}^m\frac{\partial b_{ij}}{\partial x_\ell}(X(t))b_{\ell k}(X(t))I_{kj},$

where

$I_{kj} = \int_t^{t+h}[W_k(u) - W_k(t)]\,dW_j(u).$

For $k = j$, $I_{jj} = \frac{1}{2}(\Delta W_j)^2 - \frac{1}{2}h$. For $k \neq j$, the mixed integrals have no comparable simple formula; their differences are Levy area terms, and sampling them is difficult.

5.2 Lamperti Transform

Definition

For $dX_t = b(X_t)\,dt + \sigma(X_t)\,dW_t$ with $\sigma(x) > 0$, define

$Y_t = F(X_t), \qquad F(x) = \int\frac{1}{\sigma(x)}\,dx.$

Then F is increasing and $X_t = F^{-1}(Y_t)$. Since

$F'(x) = 1/\sigma(x), \qquad F''(x) = -\sigma'(x)/\sigma^2(x),$

Ito's formula gives

$dY_t = \left[\frac{b(X_t)}{\sigma(X_t)} - \frac{1}{2}\sigma'(X_t)\right]dt + dW_t.$

Writing $X_t = F^{-1}(Y_t)$,

$dY_t = a(Y_t)\,dt + dW_t, \qquad a(y) = \frac{b(F^{-1}(y))}{\sigma(F^{-1}(y))} - \frac{1}{2}\sigma'(F^{-1}(y)).$

Thus Lamperti converts state-dependent volatility into constant volatility one. Euler through Lamperti uses

$\hat{Y}_{t_{i+1}} = \hat{Y}_{t_i} + a(\hat{Y}_{t_i})\Delta t_{i+1} + \sqrt{\Delta t_{i+1}}Z_{i+1}, \qquad \hat{X}_{t_{i+1}} = F^{-1}(\hat{Y}_{t_{i+1}}).$

GBM transform

For $dX_t = rX_t\,dt + \sigma X_t\,dW_t$,

$F(x) = \frac{1}{\sigma}\log x, \qquad Y_t = \frac{1}{\sigma}\log X_t, \qquad X_t = e^{\sigma Y_t}.$

Ito's formula gives

$dY_t = \left(\frac{r}{\sigma} - \frac{\sigma}{2}\right)dt + dW_t.$

Euler on Y is exact, so transforming back gives a scheme for X without discretization error.

5.3 Numerical Examples

Exact grid for deterministic volatility

For a discretely monitored floating-strike lookback call with payoff $S_T - \min_{i=1,\dots,m}S_{t_i}$ and

$dS_t = rS_t\,dt + \theta(t)S_t\,dW_t,$

set $Y_t = \log S_t$. Then

$dY_t = (r - \frac{1}{2}\theta^2(t))dt + \theta(t)dW_t.$

Let $\sigma_i^2 = \int_{t_i}^{t_{i+1}}\theta^2(t)\,dt$. The exact grid recursion is

$Y_{i+1} = Y_i + r\Delta t_i - \frac{1}{2}\sigma_i^2 + \sigma_iZ_{i+1}, \qquad S_{i+1} = e^{Y_{i+1}}.$

The discounted output is $H = e^{-rT}(S_m - \min(S_1,\dots,S_m))$. For $\theta(t) = \sqrt{a^2 + b^2\sin(2\pi t/T)}$,

$\sigma_i^2 = a^2\Delta t_i + \frac{b^2T}{2\pi}\{\cos(2\pi t_i/T) - \cos(2\pi t_{i+1}/T)\}.$

Exact grid simulation removes process discretization error, but a continuously monitored payoff can still be biased because the between-grid minimum is approximated.

Artificial-dynamics control

For $dS_t = rS_t\,dt + \theta_tS_t\,dW_t$ and $d\theta_t = a(\Theta - \theta_t)dt + \beta dB_t$, with $\mathrm{corr}(dW, dB) = \rho$, set $Y_t = \log S_t$. Euler gives

$\hat{Y}_{i+1} = \hat{Y}_i + (r - \frac{1}{2}\hat{\theta}_i^2)\Delta t_i + \hat{\theta}_i\sqrt{\Delta t_i}Z_{i+1},$

$\hat{\theta}_{i+1} = \hat{\theta}_i + a(\Theta - \hat{\theta}_i)\Delta t_i + \beta\sqrt{\Delta t_i}R_{i+1}.$

The call output is $X = e^{-rT}(e^{\hat{Y}_T} - K)^+$. Use the same Z_i to define an artificial GBM log process

$\overline{Y}_{i+1} = \overline{Y}_i + (r - \frac{1}{2}\sigma^2)\Delta t_i + \sigma\sqrt{\Delta t_i}Z_{i+1}.$

Then $V = e^{-rT}(e^{\overline{Y}_T} - K)^+$ has expectation $\mathrm{BSCall}(S_0, K, T, r, \sigma)$, giving the control output

$X - b\{V - \mathrm{BSCall}(S_0, K, T, r, \sigma)\}.$

Choosing σ near the long-run mean Θ gives strong variance reduction in the lecture example.

Heston with cross-entropy IS

For the Heston model

$dS_t = rS_t\,dt + \sqrt{\theta_t}S_t\,dW_t, \qquad d\theta_t = a(b - \theta_t)dt + \sigma\sqrt{\theta_t}\,dB_t,$

use the nonnegative Euler scheme

$\hat{S}_{i+1} = \hat{S}_i + r\hat{S}_i\Delta t_i + \hat{S}_i\sqrt{\hat{\theta}_i\Delta t_i}Z_{i+1},$

$\hat{\theta}_{i+1} = \max\{0, \hat{\theta}_i + a(b - \hat{\theta}_i)\Delta t_i + \sigma\sqrt{\hat{\theta}_i\Delta t_i}R_{i+1}\},$

where $R_{i+1} = \rho Z_{i+1} + \sqrt{1-\rho^2}Y_{i+1}$. The payoff is $h(X) = e^{-rT}(\hat{S}_T - K)^+$ for $X = (Z_1,\dots,Z_m,Y_1,\dots,Y_m)$.

For IS use $N(\mu, I_{2m})$, $\mu = (v, v)$. An initial tilt is

$v^0 = x(\sqrt{\Delta t_1},\dots,\sqrt{\Delta t_m}), \qquad v^0 = -\frac{\rho}{\sqrt{1-\rho^2}}v^0,$

where matching $E[S_T] \approx K$ after replacing θ_t by b gives

$x = \frac{1}{\sqrt{bT}}\log\left(\frac{K}{S_0}\right) - \frac{r}{\sqrt{b}}.$

The cross-entropy update is

$\hat{\mu}^{j+1} = \frac{\sum_{k=1}^N h(\overline{X}_k)e^{-\langle\mu^j, \overline{X}_k\rangle}\overline{X}_k}{\sum_{k=1}^N h(\overline{X}_k)e^{-\langle\mu^j, \overline{X}_k\rangle}}, \qquad \overline{X}_k \sim N(\mu^j, I_{2m}).$

After $\hat{\mu} = (\hat{v}, \hat{v})$ is chosen, simulate $Z_i \sim N(\hat{v}_i, 1)$ and $Y_i \sim N(\hat{v}_i, 1)$ and multiply the payoff by

$\exp\left\{-\sum_{i=1}^m(\hat{v}_iZ_i + \hat{v}_iY_i) + \frac{1}{2}\sum_{i=1}^m(\hat{v}_i^2 + \hat{v}_i^2)\right\}.$

The variance reduction is strongest for high strikes.

6 Mean-Risk Opt and EU Max

Denote by $R = (R_1,\dots,R_n)^\top$ the net return vector of the stocks and r_f the risk-free net return rate. Given portfolio $w = (w_1,\dots,w_n)^\top$, where w_i represents the percentage of initial wealth invested in stock i, the net return rate of the portfolio is:

$$R_w = r_f + w^\top(R - r_f1_n)$$

6.1 Mean-Semi-variance Model

One alternative to variance is semi-variance:

$$\mathbb{E}[(\min(R_w - \mathbb{E}[R_w], 0)^2)]$$

This measures the downside risk of the portfolio. If the distribution is symmetrical, semi-variance is proportional to variance; in general, it is not the case. The Mean-semi-variance portfolio selection problem is formulated as:

$$\begin{aligned} &\min_w \mathbb{E}[(\min(R_w - \mathbb{E}[R_w], 0)^2)] \\ &\text{s.t.} \quad \mathbb{E}[R_w] \geq z \end{aligned}$$

6.2 Mean-Absolute Deviation Model

Another alternative to variance is absolute deviation:

$$\mathbb{E}[|R_w - \mathbb{E}[R_w]|]$$

Mean-absolute deviation portfolio selection is formulated as:

$$\begin{aligned} &\min_w \mathbb{E}[|R_w - \mathbb{E}[R_w]|] \\ &\text{s.t.} \quad \mathbb{E}[R_w] \geq z \end{aligned}$$

6.3 Safety-First Model

Safety-first portfolio selection is formulated as:

$$\begin{aligned} &\min_w \mathbb{P}(R_w \leq D) \\ &\text{s.t.} \quad \mathbb{E}[R_w] \geq z \end{aligned}$$

Here, D is a given threshold below which the portfolio return is considered to be a disaster. Both the mean-semi-variance and the mean-absolute deviation portfolio selection problems are convex programs, but the safety-first portfolio selection problem is not.

6.4 Tail Risk Measures

In the safety-first criterion, investors are concerned about the left tail of the return distribution.

Given initial wealth x_0 and a portfolio w, the wealth at the end of the period is $X_T = x_0(1 + R_w)$, and thus the P&L of the portfolio is:

$$Z := X_T - x_0 = x_0R_w$$

Definition of VaR

Let Z be a random P&L, with negative values referring to losses and positive values to gains. Given a level $\alpha \in (0,1)$, informally the α-VaR, denoted as $VaR_\alpha(Z)$, is defined by:

$$\mathbb{P}(-Z \leq VaR_\alpha(Z)) = \alpha$$

Formally, $VaR_\alpha(Z)$ is defined as follows:

$$\begin{aligned} VaR_\alpha(Z) &= \inf\{x \in \mathbb{R} : \mathbb{P}(-Z \leq x) \geq \alpha\} \\ &= -\sup\{y \in \mathbb{R} : \mathbb{P}(Z < y) \leq 1 - \alpha\} \end{aligned}$$

Definition of CVaR

Given a level $\alpha \in (0,1)$, informally the α-CVaR, denoted as $CVaR_\alpha(Z)$, is defined by:

$$CVaR_\alpha(Z) = \mathbb{E}[-Z| - Z \geq VaR_\alpha(Z)]$$

Formally, $CVaR_\alpha(Z)$ is defined by:

$$CVaR_\alpha(Z) = \frac{1}{1-\alpha}\int_\alpha^1 VaR_\beta(Z)d\beta$$

CVaR can be derived by solving an optimization problem:

$$CVaR_\alpha(Z) = \min_b\{b + \frac{1}{1-\alpha}\mathbb{E}[\max(-Z - b, 0)]\}$$

6.5 Coherent Risk Measures

A risk measure $\rho(Z)$ assigns a numerical value to a random P&L Z. A risk measure ρ is a coherent risk measure, if it satisfies the following properties:

Monotonicity: for any Z_1 and Z_2 with $Z_1 \leq Z_2$, $\rho(Z_1) \geq \rho(Z_2)$.

Translation invariance: for any constant m and any Z, $\rho(Z + m) = \rho(Z) - m$.

Positive homogeneity: for any Z and any positive constant λ, $\rho(\lambda Z) = \lambda\rho(Z)$.

Subadditivity: for any P&Ls Z_1 and Z_2, $\rho(Z_1 + Z_2) \leq \rho(Z_1) + \rho(Z_2)$.

CVaR is a coherent risk measure. VaR satisfies monotonicity, translation invariance, and positive homogeneity, but not subadditivity; thus, it is not a coherent risk measure.

6.6 Data-Driven Mean-Risk Portfolio Selection

Suppose that we have observed T samples of R, denoted as R_t, $t = 1,\dots,T$. Assume that the future scenarios are confined to be within these T observed samples, and each has a probability of $1/T$ to occur. This means the distribution of the return rate in the next period is the empirical distribution. Denote the sample mean as:

$$\overline{R} := \frac{1}{T}\sum_{t=1}^T R_t$$

Optimization: Mean-Semi-Variance Portfolio Selection

The data-driven mean-semi-variance portfolio selection can be formulated as the following quadratic program(no borrowing and no shorting.):

$$\begin{aligned} \min_{w,y_1,\dots,y_T} \quad & \frac{1}{T}\sum_{t=1}^T y_t^2 \\ \text{s.t.} \quad & y_t \geq w^\top(\overline{R} - R_t), \quad t = 1,\dots,T \\ & y_t \geq 0, \quad t = 1,\dots,T \\ & r_f + w^\top(R - r_f1_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1,\dots,n \end{aligned}$$

Optimization: Mean-Absolute Standard Deviation Portfolio Selection

The data-driven mean-absolute standard deviation portfolio selection can be formulated as the following linear program(no borrowing and no shorting):

$$\begin{aligned} \min_{w,y_1,\dots,y_T} \quad & \frac{1}{T}\sum_{t=1}^T y_t \\ \text{s.t.} \quad & y_t \geq w^\top(R_t - \overline{R}), \quad t = 1,\dots,T \\ & y_t \geq -w^\top(R_t - \overline{R}), \quad t = 1,\dots,T \\ & r_f + w^\top(\overline{R} - r_f1_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1,\dots,n \end{aligned}$$

Optimization: Mean-CVaR Portfolio Selection

Suppose the initial wealth is x_0, so for portfolio w, the P&L is $Z_w = x_0R_w$. Suppose the target for the expected return rate is z. The data-driven mean-α-CVaR portfolio selection can be formulated as the following linear program:

$$\begin{aligned} \min_{w,y_1,\dots,y_T,b} \quad & b + \frac{1}{(1-\alpha)T}\sum_{t=1}^T y_t \\ \text{s.t.} \quad & y_t \geq -(r_f + w^\top(R_t - r_f1_n))x_0 - b, \quad t = 1,\dots,T \\ & y_t \geq 0, \quad t = 1,\dots,T \\ & r_f + w^\top(\overline{R} - r_f1_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1,\dots,n \end{aligned}$$

6.7 Expected Utility

Expected utility (EU) theory suggests that when making decisions under risk, people do not evaluate monetary payoffs directly. Instead, they attribute a subjective value (utility) to each monetary payoff. X is evaluated as the expectation of its utility $\mathbb{E}[u(X)]$. With a logarithmic utility $u(x) = \log(x)$, the expected utility of the St. Petersburg game is:

$$\mathbb{E}[u(X)] = \sum_{n=1}^{\infty}(1/2^n)\times\log(2^n) = \log\ 2\times\sum_{n=1}^{\infty} n/2^n = \log(4)$$

This solves the paradox.

Risk Aversion

Individuals are risk averse if any random payoff X is less preferred to its mean $\mathbb{E}[X]$. An individual with EU preferences is risk averse if and only if their utility function u is concave. Concavity is equivalent to diminishing sensitivity, meaning the marginal utility $u'(x)$ is decreasing.

Certainty Equivalent

The certainty equivalent of a random payoff X is the deterministic payoff c that is indifferent to X:

$$u(c) = \mathbb{E}[u(X)]$$

Individual A is more risk averse than B if the certainty equivalent of X for A is less than or equal to that for B.

Risk Aversion Coefficient

The Arrow-Pratt absolute risk aversion coefficient is defined as:

$$-\frac{u''(x)}{u'(x)}$$

The larger the coefficient, the more concave the utility function and the more risk averse the individual.

Equivalent Utility Functions

Preferences remain the same if we replace $u(x)$ with $au(x)+b$ where $a>0$. Thus, $v(x) := au(x)+b$ is equivalent to $u(x)$.

Examples of Utility Functions

Exponential utility ($\gamma > 0$):

$$u(x) = 1-e^{-\gamma x}, \quad x\in\mathbb{R}$$

Power utility ($\gamma > 0, \gamma\neq 1$):

$$u(x) = x^{1-\gamma}/(1-\gamma), \quad x>0$$

Logarithmic utility:

$$u(x) = \ln x, \quad x>0$$

Quadratic utility ($\gamma>0$):

$$u(x) = x-\frac{\gamma}{2}x^2, \quad x\in\mathbb{R}$$

6.8 Elicit Utility Function

Certainty Equivalent Method

Select reference points A and B. Propose lottery $(A,p;B,1-p)$. Ask for certainty equivalent c. Then:

$$u(c) = pu(A)+(1-p)u(B)$$

$u(A)$ and $u(B)$ are set exogenously due to scaling freedom. Varying p maps the curve.

Probability Equivalent Method

Select reference points A and B. For a value x, ask for the probability p such that the individual accepts the lottery in place of x. Then:

$$u(x) = pu(A)+(1-p)u(B)$$

Tradeoff Method

Compare lotteries $(X,p;r)$ and $(x,p;R)$ with reference outcomes $R>r$. Vary X until indifference. Then compare $(Y,p;r)$ and $(y,p;R)$. By EUT:

$$u(X)-u(x) = u(Y)-u(y)$$

This establishes an indifference relation. Starting from a base x_0, one can identify a sequence where $u(x_j) = ju(x_1)$.

6.9 Portfolio Selection with EU in Single Period

Consider a single period problem with initial wealth x_0, assets with returns R_i, and portfolio weights w_i. Portfolio return is $R_w = \sum_{i=1}^n w_iR_i$. Terminal wealth is $x_0(1+R_w)$. The objective is:

$$\begin{aligned}&\max\mathbb{E}[u(x_0(1+R_w))]\\ &\text{s.t.}\quad \sum_{i=1}^n w_i = 1\end{aligned}$$

Relation with Mean-Variance Criterion

For quadratic utility $u(x) = x-(\gamma/2)x^2$, the expected utility depends on $\mathbb{E}[R_w]$ and $Var[R_w]$. Thus, the optimal portfolio must be in the mean-variance set. Expected utility of terminal wealth:

$$x_0\left\{\mathbb{E}[1+R_{\mathbf{w}}]-(\gamma x_0/2)\mathbb{E}[(1+R_{\mathbf{w}})^2]\right\}$$

$$= x_0\left\{1-(\gamma x_0/2)+(1-\gamma x_0)\mathbb{E}[R_{\mathbf{w}}]+(\gamma x_0/2)(\mathbb{E}[R_{\mathbf{w}}])^2-(\gamma x_0/2)\mathrm{Var}[R_{\mathbf{w}}]\right\}$$

For exponential utility $u(x) = -e^{-\gamma x}$ and normally distributed returns, the expected utility is:

$$-\exp\{-\gamma x_0(1+\mathbb{E}[R_w])+\frac{\gamma}{2}x_0^2 Var(R_w)\}$$

Thus, the optimal portfolio must be on the mean-variance efficient frontier.

General Case

In general, EU and mean-variance criteria differ. Using the Lagrange dual method, the optimal portfolio w^* satisfies:

$$\begin{aligned}&\mathbb{E}[u'(x_0(1+R_{w^*}))x_0R_i] = \lambda, \quad i=1,\dots,n\\ &\text{s.t.}\quad \sum_{i=1}^n w_i^* = 1\end{aligned}$$

7 CAPM and Arbitrage Pricing Theory

7.1 Market Equilibrium and CAPM model

Market equilibrium: at equilibrium price

All investors have achieved their goals. Market is clear, demand=supply.

Capital asset pricing model (CAPM): an equilibrium pricing model in framework of mean-variance analysis.

Assumptions

Each investor is a price-taker and mean-variance optimizer (although with different risk appetite). Everyone agrees on the probabilistic structure of assets (mean, covariance matrix of future asset prices). There is a unique risk-free rate r_f of borrowing and lending that is available to all. Market is frictionless: assets can be traded in any fraction and there is no transaction fee and other fees. The equilibrium price exists.

Market Portfolio

Suppose there are n risky assets in the market with number of shares $N_i, i=1,\dots,n$. The current prices are $P_i, i=1,\dots,n$.

Total capital: $\sum_{i=1}^n N_iP_i$

Capitalization weight: $\tilde{w}_M = (\tilde{w}_{M,1},\dots,\tilde{w}_{M,n})$

$$\tilde{w}_{M,i} = \frac{\textit{capital in asset } i}{\textit{total capital}} = \frac{N_iP_i}{\sum_{i=1}^n N_iP_i}, i=1,\dots,n$$

Market portfolio: portfolio with capitalization weight.

Market Portfolio and Tangent Portfolio

Each investor holds his optimal portfolio which is a mixture of risk-free asset and mutual tangent portfolio. The total assets held by all investors must be the mixture of risk-free asset and tangent portfolio. Under equilibrium, total holding of assets equals total supply. Thus tangent portfolio is market portfolio.

Suppose there are J investors in the market. Investor j has initial wealth x_j and invests in the portfolio $(w_{j,0},(1-w_{j,0})\tilde{w}_p)$, $j=1,\dots,J$. The number of shares of the i-th risky asset investor j holds is $x_j(1-w_{j,0})\tilde{w}_{p,i}/P_i, i=1,\dots,n$.

In equilibrium:

$$(\sum_{j=1}^J x_j(1-w_{j,0}))\tilde{w}_{p,i}/P_i = N_i$$

Thus, $\sum_{j=1}^J x_j(1-w_{j,0}) = \sum_{i=1}^n N_iP_i$, and for $i=1,\dots n$:

$$\tilde{w}_{p,i} = \frac{N_iP_i}{\sum_{j=1}^J x_j(1-w_{j,0})} = \frac{N_iP_i}{\sum_{i=1}^n N_iP_i} = \tilde{w}_{M,i}, \quad i=1,\dots,n$$

Capital Market Line

The efficient set is a single straight line starting from the risk-free point and passing through the market portfolio. This line is called capital market line. Mathematically, capital market line states for any efficient portfolio of assets, its return R satisfies:

$$\mathbb{E}[R] = r_f+\frac{\mathbb{E}[R_M]-r_f}{\sigma_M}\sigma$$

where $\mathbb{E}[R]$ and σ are the mean and standard deviation of R, respectively, and $\mathbb{E}[R_M]$ and σ_M are the mean and standard deviation of the return of the market portfolio.

A short proof: Efficient portfolio $(w_0,(1-w_0)\tilde{w}_M)$; its expected return is:

$$\mathbb{E}[R] = w_0r_f+(1-w_0)\mathbb{E}[R_M] = r_f+(\mathbb{E}[R_M]-r_f)(1-w_0)$$

and its deviation is:

$$\sigma = |1-w_0|\sigma_M$$

Thus, we obtain the capital market line (assuming $\bar{r}_M\geq r_f$). Capital market line is the efficient frontier.

Derivation of Asset Pricing

Consider asset/portfolio i with return R_i, and denote its expectation and standard deviation as $\mathbb{E}[R_i]$ and σ_i, respectively. Consider the mean-variance problem with $z = \mathbb{E}[R_i]$. The optimal portfolio is:

$$\begin{bmatrix} w_0\\ (1-w_0)\tilde{w}_M\end{bmatrix}$$

with $1-w_0 = \frac{\mathbb{E}[R_i]-r_f}{\mathbb{E}[R_M]-r_f}$.

For any $\varepsilon\in\mathbb{R}$, consider the portfolio which is mixture of $(1-\varepsilon)$ of the optimal portfolio and ε of asset i. This new portfolio is feasible with expected return z and its standard deviation is:

$$\begin{aligned}&Var((1-\varepsilon)w_0r_f+(1-\varepsilon)(1-w_0)R_M+\varepsilon R_i)\\ &=(1-\varepsilon)^2(1-w_0)^2Var(R_M)+2\varepsilon(1-\varepsilon)(1-w_0)cov(R_i,R_M)+\varepsilon^2Var(R_i)\\ &=: f(\varepsilon)\end{aligned}$$

0 is the minimum point of $f(\cdot)$ because $(w_0;(1-w_0)\tilde{w}_M)$ is in the minimum-variance set.

Capital Asset Pricing Model (CAPM)

First-order condition:

$$0 = f'(0) = -2(1-w_0)^2Var(R_M)+2(1-w_0)cov(R_i,R_M)$$

leads to:

$$1-w_0 = \frac{cov(R_i,R_M)}{Var(R_M)}$$

Capital asset pricing model (CAPM):

$$\mathbb{E}[R_i]-r_f = \beta_i(\mathbb{E}[R_M]-r_f)$$

where

$$\beta_i = \frac{cov(R_i,R_M)}{Var(R_M)}$$

$$\mathbb{E}[R]-r_f = \beta(\mathbb{E}[R_M]-r_f)$$

draws security market line on $(\beta,\mathbb{E}[R])$ plane.

β_i is referred to as the beta of the asset. $\mathbb{E}[R_i]-r_f$ is termed as expected excess rate of return of asset i; $\mathbb{E}[R_M]-r_f$ is the expected excess rate of return of market portfolio.

CAPM says expected excess return of an asset is proportional to expected excess return of market portfolio. The proportion is its beta (β).

$\beta>0$, i.e., positively correlated; positive expected excess return.

$\beta=0$, i.e., uncorrelated; zero expected excess return.

$\beta<0$, i.e., negatively correlated; negative expected excess return.

It is β rather than σ that contributes to expected excess return.

Economic Interpretation of CAPM

Consider the portfolio consisting of market portfolio, ε of asset i, and $-\varepsilon$ of risk-free asset, the last two components being a zero cost trading strategy with P&L $\varepsilon(R_i-r_f)$. The variance of the portfolio is:

$$Var(R_M)+2\varepsilon cov(R_i,R_M)+\varepsilon^2Var(R_i)$$

When $\varepsilon << 1$, second term dominates third term. If $cov(R_i,R_M)=0$, asset i does not affect the variance, just like risk-free asset; thus it has zero risk premium. If $cov(R_i,R_M)>0$, asset i increases the variance; thus it has positive risk premium. If $cov(R_i,R_M)<0$, asset i reduces the variance; thus it has negative risk premium.

Key insight: risk is the variance of the portfolio return, not the variance of any single asset's return.

Decompose the Risk

For any asset i with (random) return R_i, define ε_i via:

$$R_i-r_f = \beta_i(R_M-r_f)+\varepsilon_i$$

By CAPM, $\mathbb{E}[\varepsilon_i]=0$. By the definition of β_i:

$$\begin{aligned}cov(R_M,\varepsilon_i) &= cov(R_M,R_i-r_f-\beta_i(R_M-R_f))\\ &= cov(R_M,R_i-r_f)-cov(R_M,\beta_iR_M-\beta_ir_f)\\ &= cov(R_M,R_i)-\beta_icov(R_M,R_M) = 0\end{aligned}$$

The risky return R_i can be decomposed into two parts: the part comoves with market return and the part uncorrelated with market return. As a result:

$$\sigma_i^2 = cov(\beta_i(R_M-r_f)+\varepsilon_i,\beta_i(R_M-r_f)+\varepsilon_i) = \beta_i^2\sigma_M^2+Var(\varepsilon_i)$$

The first term, $\beta_i^2\sigma_M^2$ is called systematic risk; The second term, $Var(\varepsilon_i)$, is called nonsystematic, idiosyncratic, or specific risk. Only the systematic risk contributes to expected return according to CAPM. Any asset or portfolio on capital market line (efficient frontier) has only systematic risk. Any asset or portfolio below capital market line has nonsystematic risk.

CAPM as a Pricing Formula

Certainty equivalent pricing formula: The price P of an asset with payoff Q is:

$$P = \frac{1}{1+r_f}\left[\mathbb{E}[Q]-\frac{cov(Q,R_M)(\mathbb{E}[R_M]-r_f)}{Var(R_M)}\right]$$

Pricing form of CAPM: The price P of an asset with payoff Q (random variables) is:

$$P = \frac{\mathbb{E}[Q]}{1+r_f+\beta(\mathbb{E}[R_M]-r_f)}$$

First formula is used when the correlation of future payoff with market portfolio is known. Second formula is used when β is known a priori.

Example: The Oil Venture

Example 7.6 (The oil venture) Consider the possiblity of investing in a share of a certain oil well that will produce a payoff that is random. The expected payoff is \$1,000 and the standard deviation of return is a relatively high 40%. The beta of the asset is $\beta = .6$. The risk-free rate is $r_f = 10\%$, and the expected return of the market portfolio is .17. What is the value of this share of the oil venture, based on CAPM? From the pricing formula, we have:

$$P = \frac{\mathbb{E}[Q]}{1+r_f+\beta(\mathbb{E}[R_M]-r_f)} = \frac{\$1{,}000}{1+0.1+0.6\times(0.17-0.1)} = \$876$$

Example: ABC Fund Analysis

Example 7.4 (ABC fund analysis) The ABC mutual fund has a 10-year record of rates of return. We would like to evaluate this fund's performance in terms of mean-variance portfolio theory and the CAPM.

Step 1 Estimate $\mu := \mathbb{E}[R]$, r_f, $\mu_M := \mathbb{E}[R_M]$, and β from the data:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^n R^i, \quad \hat{\mu}_M = \frac{1}{n}\sum_{i=1}^n R_M^i, \quad \hat{r}_f = \frac{1}{n}\sum_{i=1}^n r_f^i$$

$$\widehat{cov(R,R_M)} = \frac{1}{n-1}\sum_{i=1}^n(R^i-\hat{\mu})(R_M^i-\hat{\mu}_M)$$

$$\widehat{Var(R_M)} = \frac{1}{n-1}\sum_{i=1}^n(R_M^i-\hat{\mu}_M)^2$$

$$\hat{\beta} = \frac{\widehat{cov(R,R_M)}}{\widehat{Var(R_M)}}$$

Step 2 We write the formula:

$$\hat{\mu}-\hat{r}_f = J+\hat{\beta}(\hat{\mu}_M-\hat{r}_f)$$

If ABC fund is priced according to CAPM, J should be zero statistically. J is known as Jensen's index or alpha (α). For ABC fund, $J>0$. If you are a CAPM believer, you may think ABC fund is underpriced and thus a good asset. If you are market efficiency believer, you may think CAPM is invalid.

Step 3 Find the estimate of the Sharpe ratio:

$$S := \frac{\hat{\mu}-\hat{r}_f}{\hat{\sigma}}$$

where

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^n(R^i-\hat{\mu})^2$$

For ABC fund, $S = 0.436$. For S&P 500, $S=0.467$. If you are a CAPM believer, you may think ABC fund is not efficient.

Testing CAPM Using Linear Regression

Suppose there are n risky assets in the market with return rates in the following period to be R_i, $i=1,\dots,n$. Suppose the returns of the risk-free asset and of the market portfolio in the following period are r_f and r_M, respectively.

Decomposition:

$$R_i-r_f = \alpha_i+\beta_i(R_M-r_f)+\varepsilon_i, \quad i=1,\dots,n$$

where ε_i are independent of each other and of r_M.

Vector form:

$$\mathbf{R}-r_f\mathbf{1}_n = \boldsymbol{\alpha}+\boldsymbol{\beta}(R_M-r_f)+\boldsymbol{\varepsilon}$$

where $\mathbf{R} := (R_1,\dots,R_n)^\top$, $\boldsymbol{\alpha} := (\alpha_1,\dots,\alpha_n)^\top$, $\boldsymbol{\beta} := (\beta_1,\dots,\beta_n)^\top$, and $\boldsymbol{\varepsilon} := (\varepsilon_1,\dots,\varepsilon_n)^T$. $\boldsymbol{\varepsilon}$ is independent of R_M.

CAPM holds if and only if $\boldsymbol{\alpha}=0$. To test whether CAPM holds, suppose we have historical data of the asset returns: R_t, $R_{M,t}$, $r_{f,t}$, $t=1,\dots,T$. Suppose the asset returns in different periods are i.i.d. Run vector linear regression to estimate α and β. Reject CAPM if the estimate of α is significantly different from zero.

7.2 Arbitrage Pricing Theory

A Multi-Factor Model

Fix one period and consider n zero-cost strategies in this period with payoffs $\tilde{r}_1,\dots,\tilde{r}_n$, respectively. Assume the following factor model:

$$\tilde{r}_i = \alpha_i+\sum_{k=1}^m\beta_{i,k}f_k+\varepsilon_i, \quad i=1,\dots,n$$

where f_k, $k=1,\dots,m$ are m common factors that are payoffs of m zero-cost strategies. ε_i represent idiosyncratic risk, i.e., $\mathbb{E}[\varepsilon_i]=0$, $cov(\varepsilon_i,f_k)=0$, and $cov(\varepsilon_i,\varepsilon_j)=0\ i=1,\dots,n,\ j=1,\dots,n\ k=1,\dots,m$, and $i\neq j$. $\beta_{i,k}$ is the loading of asset i on factor k. α_i are constants.

Example: CAPM ($m=1$). $f=R_M-r_f$, $\tilde{r}_i = R_i-r_f$, $\alpha_i=0$.

Example: Fama-French three-factor model ($m=3$). $f_1 = R_M-r_f$. f_2, usually denoted as SMB, is the difference in the returns of two portfolios that are related to the so-called size premium. f_3, usually denoted as HML, is the difference in the returns of two portfolios that are related to the so-called value premium.

Vector form of the factor model:

$$\tilde{\mathbf{r}} = \boldsymbol{\alpha}+\boldsymbol{\beta}\mathbf{f}+\boldsymbol{\varepsilon}$$

where $\mathbf{f} := (f_1,\dots,f_m)^\top$, $\boldsymbol{\beta} := \{\beta_{i,j}\}$, $\boldsymbol{\varepsilon} := (\varepsilon_1,\dots,\varepsilon_n)^T$, $\boldsymbol{\alpha} := (\alpha_1,\dots,\alpha_n)^\top$.

Arbitrage Pricing Theory (APT)

Theorem (Arbitrage Pricing Theory (APT)): Assume there is no arbitrage opportunity in the market. Then, $\alpha=0$. In consequence,

$$\mathbb{E}[\tilde{r}] = \beta\mathbb{E}[f]$$

$\mathbb{E}[f]$ is the risk premium of the common factors. The risk premium of asset i is its loading on the common factors multiplied by the risk premia of the common factors.

In CAPM, the only factor is the excess return of the market portfolio, R_M-r_f. The risk premium of the common factor is the expected excess return of the market portfolio, $\mathbb{E}[R_M]-r_f$. The loading of asset i on the common factor is its beta.

Statistical Testing of APT

Suppose we have historical data of the payoffs of the n zero-cost strategies: $\tilde{r}_t$, $t=1,\dots,T$ and the historical data of the payoffs of the m factors: $f_t, t=1,\dots,T$. Suppose the payoffs are i.i.d. across periods. Run multiple linear regression to estimate α and β. Reject APT if the estimate of α is significantly different from zero.

8 Multi-Period Investment Models

8.1 Multi-Period Investment

There are N periods of investment. Denote the beginning and end of j-th period as t_{j-1} and t_j, $j=1,\dots,N$. There is a risk-free asset and m risky stocks in the market. The net return rate of investing in the risk-free asset in the j-th period is a constant $r_{f,j}$. The excess return rate of investing in stock i in period j is $R_{i,j}$.

Assume that $R_j := (R_{1,j},\dots,R_{m,j})^\top$, $j=1,\dots,N$ are independent.

Consider an agent with initial wealth x_0 at the beginning of first period. The agent's decision is the allocation of his money in each period.

8.2 Wealth Dynamics

Denote as X_j the wealth of the agent at time t_j, $j=0,1,\dots,N$. Suppose that except for the initial investment, the agent does not inject new money nor withdraw money from the investment account before the terminal time $T := t_N$.

At the beginning of each period j, the agent decides to invest $\pi_{i,j-1}$ dollars in stock i, $i=1,\dots,m$, and the remaining amount of money in the risk-free asset. Wealth dynamics:

$$X_j = (1+r_{f,j})X_{j-1}+\sum_{i=1}^m\pi_{i,j-1}R_{i,j}, \quad j=1,\dots,N$$

8.3 Expected Utility Maximization

Assume that the agent maximizes the expected utility of her terminal wealth,

$$\max_{\pi_{j-1},j=1,\dots,N}\mathbb{E}[u(X_N)]$$

where u is the agent's utility function and $\pi_{j-1} = (\pi_{1,j-1},\dots,\pi_{m,j-1})^\top$, $j=1,\dots,N$ are the agent's portfolio strategy.

8.4 Dynamic Programming

We apply dynamic programming to solve the problem. Principle of Optimality: Starting from any point on the optimal path, the remaining decisions must constitute an optimal policy with regard to the point.

Consider time t_k when the agent's wealth then is x. Consider the following problem:

$$\max_{\pi_{j-1},j=k+1,\dots,N}\mathbb{E}_k[u(X_N)|X_k=x]$$

Denote the above optimal value function as $v_k(x)$.

Dynamic Programming (DP) Equation:

$$\begin{aligned}v_k(x) &= \max_{\pi_k}\mathbb{E}_k[v_{k+1}(X_{k+1})|X_k=x], \quad k=0,1,\dots,N-1\\ v_N(x) &= u(x)\end{aligned}$$

Instead of solving a dynamic portfolio selection problem, we solve a family of single-period portfolio selection problem. $(\pi_0^*,\dots,\pi_{N-1}^*)$ is optimal to the dynamic portfolio selection problem if and only if π_k^* is optimal to the above single-period portfolio selection problem, $k=0,\dots,N-1$.

8.5 Power Utility

In the following, we assume power utility function:

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad x>0$$

for some $0<\gamma\neq 1$. Applying DP, we have:

$$v_{N-1}(x) = \max_{\pi_{N-1}}\mathbb{E}_{N-1}[u(X_N)|X_{N-1}=x] = b_{N-1}x^{1-\gamma}, \quad x>0$$

where

$$b_{N-1} = \max_w\frac{\mathbb{E}[(1+r_f+w^\top R_N)^{1-\gamma}]}{1-\gamma}$$

Denote the optimizer in the right-hand side of the above as w_{N-1}^*.

By induction, we conclude:

$$v_{k-1}(x) = (\prod_{j=k-1}^{N-1}(b_j(1-\gamma)))\frac{x^{1-\gamma}}{1-\gamma}$$

where

$$b_{k-1} = \max_w\frac{\mathbb{E}[(1+r_f+w^\top R_k)^{1-\gamma}]}{1-\gamma}, \quad k=1,\dots,N$$

Denote the optimizer in the right-hand side of the above as w_{k-1}^*. Then, the optimal portfolio is:

$$\pi_{k-1}^* = w_{k-1}^*X_{k-1}, \quad k=1,\dots N$$

Myopic Portfolio

With the power utility function, the investor acts at each stage, based on the optimal policy, as if he were faced with a single-period investment, i.e., as if he were myopic. When the utility function is not a power function or when the stock returns are not independent over time, the above is not necessarily true.

For power utility functions, it is implicitly assumed that the wealth must remain positive. Thus, the portfolio π needs to be constrained, i.e., $\pi_{i,k}\geq 0$, $i=1,\dots,n$, and $\sum_{i=1}^n\pi_{i,k}\leq X_k$.

Example: Expected Utility Maximization

Suppose the risk-free net return rate in each period is 0.03, and there is one stock in the market only with i.i.d. excess return rates over time. Suppose that the log return rate of the stock, $\ln(1+r_f+R_k)$ follows normal distribution with mean 0.07 and variance 0.04.

Suppose that the investor is not allowed to short-sell the stock or to borrow, i.e., $0\leq\pi_k\leq X_k$. Suppose that the investor has a power utility function with $\gamma=10$.

To find the optimal portfolio, we need to solve:

$$\max_{w\in[0,1]}f(w)$$

where

$$f(w) := \frac{\mathbb{E}[(1+r_f+R_kw)^{1-\gamma}]}{1-\gamma}$$

Straightforward calculation yields:

$$f'(w) = \mathbb{E}[(1+r_f+R_kw)^{-\gamma}R_k]$$

By binary search, one can find that $f'(0.1489)=0$, which implies that the optimal portfolio $w^*=0.1489$, i.e., it is optimal to invest 14.89

8.6 Long-Run Mean Exponential Growth Rate

Suppose $t_k-t_{k-1}=\Delta$, $k=1,\dots,N$ in the above, e.g., Δ is one month, so the terminal time $t_N = N\Delta$. Assume that the stock excess returns $\mathbf{R}_k$, $k=1,2,\dots$ are i.i.d.

For each portfolio strategy $(\boldsymbol{\pi}_1,\boldsymbol{\pi}_2,\dots)$, consider the long-run mean exponential growth rate:

$$\lim_{N\to+\infty}\mathbb{E}\left[\frac{\log(X_N/X_0)}{N}\right]$$

The objective is to maximize the above long-run mean exponential growth rate.

Kelly's Portfolio

It turns out that the optimal strategy is as follows: at each time $k\Delta$ invest w_i^* proportion of wealth at that time in stock i, $i=1,\dots,n$, where

$$w^* := (w_1^*,\dots,w_n^*)^\top = \mathrm{argmax}_w\mathbb{E}[\log(1+r_f+w^\top R_k)]$$

The above strategy is called Kelly strategy.

Kelly portfolio has many interesting properties: Asymptotically maximizes the rate of asset growth. Asymptotically maximizes the median of wealth. Is a myopic strategy.

The Kelly strategy may lead to a large fraction of wealth in bets or in risky investment. For instance, for a stock with annual mean excess return rate 4% and volatility 20%, the Kelly strategy is to invest 100% of wealth in the stock.

Fractional Kelly

MacLean et al. (1992) propose fractional Kelly to tradeoff growth and security. κ-fractional Kelly refers to a strategy of investing κ fraction of one's wealth in the Kelly portfolio and holding the remaining as cash (or in the risk-free asset). Fractional Kelly is a heuristic portfolio, rather than the consequence of optimality. MacLean et al. (1992) lay out two attractive properties of fractional Kelly: It is easily computable. There is an effective tradeoff between growth and security by varying κ.

8.7 Multi-Period Mean-Variance Portfolio

Next, we consider multi-period mean-variance portfolio selection. The market setting is as before. Denote by μ_k and C_k the mean and variance of R_k, respectively.

At time 0, given initial wealth x_0, the agent considers the following dynamic portfolio selection problem:

$$\begin{aligned}&\min_{\pi_0,\dots,\pi_{N-1}}\frac{1}{2}Var(X_N)\\ &\text{s.t.}\quad \mathbb{E}[X_N]=z_0\\ &X_j = (1+r_{f,j})X_{j-1}+\pi_{j-1}^\top R_j, \quad j=1,\dots,N\\ &X_0=x_0\end{aligned}$$

where z_0 is the agent's target for expected terminal wealth.

Expected Quadratic Utility

Assume that C_k is positive definite, $k=1,\dots,N$.

Assume that $z_0>\prod_{j=1}^N(1+r_{f,j})$.

The mean-variance portfolio selection problem is equivalent to the following:

$$\begin{aligned}&\min_{\pi_0,\dots,\pi_{N-1}}\frac{1}{2}\mathbb{E}[(X_N-z_0)^2]\\ &\text{s.t.}\quad \mathbb{E}[X_N]=z_0\\ &X_j = (1+r_{f,j})X_{j-1}+\pi_{j-1}^\top R_j, \quad j=1,\dots,N\\ &X_0=x_0\end{aligned}$$

Lagrange Dual Method

Apply the Lagrange multiplier λ_0 to the constraint for the expected terminal wealth target, and we obtain:

$$\min_{\pi_0,\dots,\pi_{N-1}}\frac{1}{2}\mathbb{E}[(X_N-z_0)^2]-\lambda_0(\mathbb{E}[X_N]-z_0)$$

subject to

$$\begin{aligned}X_j &= (1+r_{f,j})X_{j-1}+\pi_{j-1}^\top R_j, \quad j=1,\dots,N\\ X_0 &= x_0\end{aligned}$$

Then, we apply dynamic programming to solve the above problem: for each $k=0,1,\dots,N-1$ and each x, consider:

$$\begin{aligned}&\min_{\pi_k,\dots,\pi_{N-1}}\frac{1}{2}\mathbb{E}_k[(X_N-z_0)^2|X_k=x]-\lambda_0(\mathbb{E}_k[X_N|X_k=x]-z_0)\\ &\text{s.t.}\quad X_j=(1+r_{f,j})X_{j-1}+\pi_{j-1}^\top R_j, \quad j=k+1,\dots,N\\ &X_k=x\end{aligned}$$

Denote the optimal value of the above as $v_k(x)$.

Dynamic Programming

DP principle:

$$\begin{aligned}v_k(x) &= \min_{\pi_k}\mathbb{E}_k[v_{k+1}((1+r_{f,k+1})X_k+\pi_k^\top R_{k+1})|X_k=x]\\ &= \min_{\pi_k}\mathbb{E}[v_{k+1}((1+r_{f,k+1})x+\pi_k^\top R_{k+1})], \quad k=0,1,\dots,N-1\\ v_N(x) &= \frac{1}{2}x^2-(z_0+\lambda_0)x+\frac{1}{2}z_0^2+\lambda_0z_0\end{aligned}$$

Denote $D_k := \mathbb{E}[R_kR_k^\top] = C_k+\mu_k\mu_k^\top$, $k=1,\dots,N$.

Optimal Portfolio

Consequently, the optimal portfolio is:

$$\boldsymbol{\pi}_k^*(x) = (1+r_{f,k+1})[(\prod_{j=k}^{N-1}(1+r_{f,j+1}))^{-1}(z_0+\lambda_0)-x]\mathbf{D}_{k+1}^{-1}\boldsymbol{\mu}_{k+1}$$

where the multiplier λ_0 is to be determined. By maximizing $v_0(x_0)$ with respect to the multiplier λ_0, we can derive the multiplier:

$$\lambda_0 = -z_0+\frac{1}{1-\prod_{j=0}^{N-1}(1-\boldsymbol{\mu}_{j+1}^\top\mathbf{D}_{j+1}^{-1}\boldsymbol{\mu}_{j+1})}$$

$$\times[z_0-(\prod_{j=0}^{N-1}(1+r_{f,j+1}))(\prod_{j=0}^{N-1}(1-\boldsymbol{\mu}_{j+1}^\top\mathbf{D}_{j+1}^{-1}\boldsymbol{\mu}_{j+1}))x_0]$$

To summarize, the optimal portfolio is:

$$\boldsymbol{\pi}_k^*(x) = (1+r_{f,k+1})\left[s_k^{-1}\left(\frac{z_0-\beta_0s_0x_0}{1-\beta_0}\right)-x\right]\mathbf{D}_{k+1}^{-1}\boldsymbol{\mu}_{k+1}$$

where s_k stands for the risk-free growth rate from the beginning of k-th period to the end, i.e., $s_k := \prod_{j=k}^{N-1}(1+r_{f,j+1})$, and $\beta_k := \prod_{j=k}^{N-1}(1-\boldsymbol{\mu}_{j+1}^\top\mathbf{D}_{j+1}^{-1}\boldsymbol{\mu}_{j+1})$.

Moreover, the optimal variance is $2v_0(x_0)$ at the optimal λ_0 and at the initial wealth x_0, i.e., the variance of the terminal wealth under the optimal strategy is:

$$Var(X_N) = \frac{\beta_0}{1-\beta_0}(z_0-s_0x_0)^2$$

The above traces out the efficient frontier as z_0 varies in $[s_0x_0,+\infty)$. The mutual fund theorem still holds.

8.8 Continuous-Time Model

Market

We can extend the multi-period model to a continuous-time model. Suppose the market has m risky stocks with prices at time t denoted as $S_{i,t}$, $i=1,\dots,m$, and a risk-free asset with price $S_{0,t}$ at time t. Suppose that the risk-free asset generates net return rate rdt in an infinitesimally small time period $[t,t+dt]$, where r is a constant, i.e.,

$$dS_{0,t}/S_{0,t} = rdt$$

Suppose that each stock i generates excess return rate $b_idt+\sum_{j=1}^m\sigma_{i,j}dW_{j,t}$ for some constants b_i and $\sigma_{i,j}$, $j=1,\dots,m$, where $(W_{1,t})_{t\geq 0},\dots,(W_{m,t})_{t\geq 0}$ are m independent standard Brownian motions. In other words,

$$dS_{i,t}/S_{i,t} = (r+b_i)dt+\sum_{j=1}^m\sigma_{i,j}dW_{j,t}$$

Portfolio

Assume that $\sigma := (\sigma_{i,j})_{i,j=1,\dots m}$ is invertible. Denote by $\pi_t = (\pi_{1,t},\dots,\pi_{m,t})^\top$ the agent's portfolio held in the infinitesimally small time period $[t,t+dt]$ i.e., $\pi_{i,t}$ dollars are invested in stock i, $i=1,\dots,m$, and the remaining is invested in the risk-free asset in this period. Then, the agent's wealth, denoted as $(X_t)_{t\geq 0}$, evolves as:

$$dX_t = rX_t dt + \pi_t(bdt + \sigma dW_t)$$

where $b := (b_1,\dots,b_m)^\top$ and $W_t := (W_{1,t},\dots,W_{m,t})^\top$. Suppose the agent's wealth at the initial time 0 is x_0.

Expected Utility Maximization

Consider maximization of expected utility of wealth at a given terminal time T:

$$\begin{aligned} &\max_{(\pi_t)_{t\in[0,T]}} \mathbb{E}[u(X_T)] \\ \text{s.t.}\quad & dX_t = rX_t dt + \pi_t^\top(bdt + \sigma dW_t) \\ & X_0 = x_0 \end{aligned}$$

where $u(\cdot)$ is the agent's utility function.

For power utility function $u(x) = x^{1-\gamma}/(1-\gamma), x>0$ for certain $0<\gamma\neq 1$, the optimal portfolio is:

$$\pi_t^* = \tfrac{1}{\gamma}(\sigma\sigma^\top)^{-1}b, \quad t\in[0,T]$$

Mean-Variance Analysis

Consider mean-variance portfolio selection. Suppose the target for the expected wealth at the terminal time T is $z_0 > x_0 e^{rT}$. The mean-variance problem is:

$$\begin{aligned} &\max_{(\pi_t)_{t\in[0,T]}} \tfrac{1}{2} Var(X_T) \\ \text{s.t.}\quad & dX_t = rX_t dt + \pi_t^\top(bdt + \sigma dW_t) \\ & X_0 = x_0 \\ & \mathbb{E}[X_T] = z_0 \end{aligned}$$

The optimal portfolio is:

$$\pi_t^* = \left[e^{-r(T-t)} \frac{z_0 - e^{-b^\top(\sigma\sigma^\top)^{-1}bT} e^{rT} x_0}{1-e^{-b^\top(\sigma\sigma^\top)^{-1}bT}} - X_t\right](\sigma\sigma^\top)^{-1}b, \quad t\in[0,T]$$

The optimal variance is:

$$Var(X_T) = \frac{e^{-b^\top(\sigma\sigma^\top)^{-1}bT}}{1-e^{-b^\top(\sigma\sigma^\top)^{-1}bT}}(z_0 - e^{rT}x_0)^2$$

9 Limit Order Book

9.1 Bid, Ask, and Mid Prices

The lowest price among all sell limit orders is referred to as ask price.
The highest price among all buy limit orders is referred to as bid price.
The difference between the bid and ask prices is referred to as bid-ask spread.
The average of the bid and ask prices is mid price.

9.2 Resolution Parameters

In general, an order $x = (p_x, \omega_x, t_x)$ submitted at time t_x with price p_x and size $\omega_x > 0$ (respectively, $\omega_x < 0$) is a commitment to sell (respectively, buy) up to $|\omega_x|$ units of the traded at a price no less than (respectively, no greater than) p_x.
The lot size σ of an LOB is the smallest amount of the asset that can be traded within it. All orders must arrive with a size $\omega_x \in \{\pm k\sigma : k = 1,2,\dots\}$.
The tick size π of an LOB is the smallest permissible price interval between different orders within it. All orders must arrive with a price that is specified to the accuracy of π.
The lot size σ and tick size π of an LOB are collectively called its resolution parameters.

9.3 Limit Order Book

An LOB $\mathscr{L}(t)$ is the set of all active orders in a market at time t. The active orders in an LOB $\mathscr{L}(t)$ can be partitioned into the set of active buy orders $\mathscr{B}(t)$, for which $\omega_x < 0$, and the set of active sell orders $\mathscr{A}(t)$, for which $\omega_x > 0$.

9.4 Depth

The bid-side depth available at price p and at time t is

$$n^b(p,t) := \textstyle\sum_{\{x\in\mathscr{B}(t):p_x=p\}} \omega_x$$

The ask-side depth available at price p and at time t, denoted $n^a(p,t)$, is defined similarly using $\mathscr{A}(t)$.
The bid price at time t is the highest stated price among all active buy orders at time t, i.e.,

$$b(t) := \max_{x\in\mathscr{B}(x)} p_x$$

The ask price at time t is the lowest stated price among all active sell orders at time t, i.e.,

$$a(t) := \min_{x\in\mathscr{A}(x)} p_x$$

9.5 Relative Price

The bid-ask spread at time t is $s(t) := a(t) - b(t)$. The mid price at time t is $m(t) := [a(t)+b(t)]/2$.
For a given price p, the bid-relative price is $\delta^b(p) := b(t) - p$ and the ask-relative price is $\delta^a(p) := p - a(t)$.
For a given order $x = (p_x, \omega_x, t_x)$, the relative price of the order is $\delta_x := \delta^b(p_x)$ if the order is a buy order, and $\delta_x := \delta^a(p_x)$ if the order is a sell order.
The bid-side depth available at relative price δ and at time t is

$$N^b(\delta,t) = \textstyle\sum_{\{x\in\mathscr{B}(t):\delta^x=\delta\}} \omega_x$$

The ask-side is defined similarly.

9.6 Price changes in LOBs

Consider a buy (respectively, sell) order $x = (p_x, \omega_x, t_x)$ that arrives immediately after time t. If $p_x \leq b(t)$ (respectively, $p_x \geq a(t)$), then x is a limit order that becomes active upon arrival. It does not cause $b(t)$ or $a(t)$ to change.
If $b(t) < p_x < a(t)$, then x is a limit order that becomes active upon arrival. Upon arrival, $b(t_x) = p_x$ (respectively, $a(t_x) = p_x$).
If $p_x \geq a(t)$ (respectively, $p_x \leq b(t)$), then x is a market order that immediately matches to one or more active sell (respectively, buy) orders upon arrival. Whenever such a matching occurs, it does so at the price of the active order, which is not necessarily equal to the price of the incoming order. Whether or not such a matching causes $a(t)$ (respectively, $b(t)$) to change at time t_x depends on $n^a(a(t),t)$ (respectively, $n^b(b(t),t)$) and ω_x.
In particular, the new bid price $b(t_x)$ upon arrival of a sell market order x is

$$b(t_x) = \max(p_x, q), \text{ where } q = \operatorname{argmax}_{k'}^{k=b(t)} |n^b(k,t)| > |\omega_x|$$

Similarly, the new ask price $a(t_x)$ upon arrival of a buy market order x is

$$a(t_x) = \min(p_x, q), \text{ where } q = \operatorname{argmin}_{k'} \textstyle\sum_{k=a(t)}^{k'} |n^a(k,t)| > |\omega_x|$$

Put another way, the incoming order x matches to the highest priority active order y of opposite type.

10 Sample Final Exam Paper

Question 1

Consider P&L Z with the following distribution:

$$\mathbb{P}(Z=-12)=0.01, \mathbb{P}(Z=-6)=0.02, \mathbb{P}(Z=-4)=0.01, \mathbb{P}(Z=-1)=0.02$$
$$\mathbb{P}(Z=0)=0.5, \mathbb{P}(Z=1)=0.24, \mathbb{P}(Z=3)=0.2$$

Compute the 95%-VaR, the 95% CVaR, the expected loss conditional on the loss being larger than or equal to the 95%-VaR, and the expected loss conditional on the loss being strictly larger than the 95%-VaR.

Solution 1

We compute the CDF of Z, denoted as $F_{-Z}(x)$. Calculation yields:

$$F_{-Z}(x) = \begin{cases} 0, & x < -3, \\ 0.2, & x\in[-3,-1), \\ 0.44, & x\in[-1,0), \\ 0.94, & x\in[0,1), \\ 0.96, & x\in[1,4), \\ 0.97, & x\in[4,6), \\ 0.99, & x\in[6,12), \\ 1, & x\geq 12. \end{cases}$$

As a result, we have:

$$VaR_\alpha(Z) = \inf\{x\in\mathbb{R} : \mathbb{P}(-Z\leq x)\geq\alpha\} = \inf\{x\in\mathbb{R} : F_{-Z}(x)\geq\alpha\}$$

Specifically for different α:

$$VaR_\alpha(Z) = \begin{cases} -3, & \alpha\in(0,0.2], \\ -1, & \alpha\in(0.2,0.44], \\ 0, & \alpha\in(0.44,0.94], \\ 1, & \alpha\in(0.94,0.96], \\ 4, & \alpha\in(0.96,0.97], \\ 6, & \alpha\in(0.97,0.99], \\ 12, & \alpha\in(0.99,1). \end{cases}$$

Consequently, $VaR_{95\%}(Z) = 1$.
The 95% CVaR is calculated as:

$$CVaR_{95\%}(Z) = \tfrac{1}{1-0.95}\int_{0.95}^1 VaR_\beta(Z)d\beta$$
$$= \tfrac{1}{0.05}\times[1\times 0.01 + 4\times 0.01 + 6\times 0.02 + 12\times 0.01] = 5.8$$

The expected loss conditional on the loss being larger than or equal to the 95%-VaR ($VaR_{95\%} = 1$) is:

$$\mathbb{E}[-Z \mid -Z \geq VaR_{95\%}(Z)] = \mathbb{E}[-Z \mid -Z \geq 1]$$
$$= \tfrac{1}{0.02+0.01+0.02+0.01}\times[1\times 0.02 + 4\times 0.01 + 6\times 0.02 + 12\times 0.01] = 5$$

The expected loss conditional on the loss being strictly larger than the 95%-VaR is:

$$\mathbb{E}[-Z \mid -Z > VaR_{95\%}(Z)] = \mathbb{E}[-Z \mid -Z > 1]$$
$$= \tfrac{1}{0.01+0.02+0.01}\times[4\times 0.01 + 6\times 0.02 + 12\times 0.01] = 7$$

Question 2

Consider a wealth manager who is looking into a particular portfolio strategy. She found that the historical estimates of the mean and variance of the portfolio's monthly return are 0.7% and 1.8%, respectively. She also estimated the mean and variance of the market portfolio's monthly return to be 0.6% and 1.2%, respectively, and the correlation between the return rates of the portfolio and the market is 0.9. The monthly risk-free return rate is flat at 0.2% in the past.

(a) According to CAPM, is this strategy a good one? Why?

(b) The director of the wealth manager believes that the Fama-French three-factor model instead of CAPM should be used to assess the portfolio strategy. To this end, the wealth manager estimated the loadings of the portfolio for SMB and HML, which turn out to be -0.3 and 0.7, respectively. She also estimated the expected payoffs for the zero-cost strategies associated with SMB and HML to be 0.4% and 0.3%, respectively. Is this strategy a good one under the Fama-French three-factor model? Why?

Solution 2

(a) We compute β_i first:

$$\beta_i = \frac{cov(r_i, r_M)}{Var(r_M)} = \frac{\rho\sigma_i\sigma_M}{\sigma_M^2} = \frac{0.9\times\sqrt{1.8\%}\times\sqrt{1.2\%}}{1.2\%} = 1.10227$$

As a result, the risk premium of the portfolio strategy implied by CAPM is:

$$\bar{r}_i - r_f = \beta(\bar{r}_M - r_f) = 1.10227\times(0.6\% - 0.2\%) = 0.44091\%$$

This is lower than the estimated risk premium, namely $0.7\% - 0.2\% = 0.5\%$. So under CAPM, this portfolio strategy is good (it offers a positive alpha).

(b) The loadings of the strategy on the market risk factor, SMB, and HML, are 1.10227, -0.3, and 0.7, respectively. The risk premiums of these three factors are 0.6%-0.2%, 0.4%, and 0.3%, respectively. Thus, the model risk premium of the strategy under the Fama-French three-factor model is:

$$1.10227\times(0.6\% - 0.2\%) - 0.3\times 0.4\% + 0.7\times 0.3\% = 0.531\%$$

This is higher than the estimated risk premium of 0.5%. Thus, this strategy is a bad one when benchmarked to the Fama-French three-factor model.

Question 3

Suppose that David's total wealth is \$40,000, in which his car is worth \$15,000. Suppose his utility function is $u(x) = 2\sqrt{x}$. There is a 5% probability that his car would be stolen and he will lose \$15,000. There is also a 10% probability that his car would be damaged in an accident and he will have to spend \$10,000 to repair it.

(a) What is the Arrow-Pratt absolute risk aversion coefficient of David?

(b) If he wants to buy a car insurance which can fully cover his loss in both cases, i.e., stolen and damaged, what is the maximum amount of money he is willing to pay for the insurance?

Solution 3

(a) The Arrow-Pratt absolute risk aversion coefficient is:

$$r(x) = -\frac{u''(x)}{u'(x)} = -\frac{2\times(1/2)\times(-1/2)x^{-3/2}}{2\times(1/2)x^{-1/2}} = \tfrac{1}{2}x^{-1}$$

(b) The maximum amount of money, A, David is willing to pay is the amount such that David is indifferent between his wealth if he purchases the insurance contract and his wealth if he does not purchase the contract.
If David purchases the insurance contract, his wealth is $40000 - A$ and the utility of the wealth is $2\sqrt{40000-A}$.
If David does not purchase the insurance contract, the expected utility of his wealth is:

$$2\sqrt{40000-15000}\times 0.05 + 2\sqrt{40000-10000}\times 0.10 + 2\sqrt{40000}\times 0.85 = 390.4524$$

Setting the expected utility of the wealth in the case in which David purchases the insurance equal to the case in which he does not purchase the insurance, namely:

$$2\sqrt{40000-A} = 390.4524$$

We derive $A = 1886.731$.

Question 4

Consider the following Limit Order Book (LOB) for a security:
Bid Side (Depth Available at Price): \$1.99 (Depth: 2), \$1.98 (Depth: 3), \$1.97 (Depth: 1), \$1.96 (Depth: 2), \$1.95 (Depth: 4).
Ask Side (Depth Available at Price): \$2.02 (Depth: 1), \$2.03 (Depth: 2), \$2.04 (Depth: 2), \$2.05 (Depth: 3), \$2.06 (Depth: 4).
Current Bid Price: \$1.99. Current Ask Price: \$2.02. Current Mid Price: \$2.005.

(a) Suppose an order arrives to buy 3 units of the security at price \$2.00. Calculate the mid price immediately after the arrival of the order.

(b) Suppose an order arrives to sell 1 unit of the security at price \$2.02. Calculate the mid price immediately after the arrival of the order.

(c) Suppose an order arrives to buy 3 units of the security at the market price. Calculate the mid price immediately after the arrival of the order.

(d) Suppose cancelation of an existing sell orders of size 1 at price \$2.02 arrives. Calculate the mid price immediately after the arrival of cancelation.

Solution 4

(a) The order is a buy limit order at \$2.00. Since \$2.00 < Ask (\$2.02), it becomes the new best bid. New Bid: \$2.00. Ask remains: \$2.02. Mid Price:

$$\frac{2.02+2.00}{2} = \$2.01$$

(b) The order is a sell limit order at \$2.02. Since \$2.02 > Bid (\$1.99) and \$2.02 is the current Ask, it adds depth to the ask side but does not change the best prices. Bid: \$1.99. Ask: \$2.02. Mid Price remains:

$$\frac{2.02+1.99}{2} = \$2.005$$

(c) A market buy order for 3 units will walk up the book. It consumes the 1 unit at \$2.02. It consumes 2 units at \$2.03. The new best Ask becomes \$2.04. Bid remains \$1.99. Mid Price:

$$\frac{2.04+1.99}{2} = \$2.015$$

(d) Cancelation of 1 unit at \$2.02. The existing depth at \$2.02 was 1. After cancelation, depth at \$2.02 becomes 0. The next available ask is \$2.03. New Ask: \$2.03. Bid remains \$1.99. Mid Price:

$$\frac{2.03+1.99}{2} = \$2.01$$

Question 5

Consider a 3-period mean-variance portfolio selection problem with one risk-free asset and 1 risky stock. Suppose the initial wealth is \$1. Suppose that the risk-free net return in each period is $r_{f,k} = 0.03$. In addition, for $k = 1,\dots,3$ the expected excess return vector of the stocks is:

$$\mu_k = \mathbb{E}[R_k] = 5\%$$

and variance of the stock excess return rate is:

$$C_k = Var(R_k) = 0.09$$

Suppose the agent's target is to grow the wealth by 6% in each period. Find the optimal portfolio strategy of the agent.

Solution 5

The number of investment periods is $N = 3$. We first compute D_k:

$$D_k = \mathbb{E}[R_k R_k^\top] = C_k + \mu_k\mu_k^\top = (5\%)^2 + 0.09 = 0.0925$$
$$\mu_k^\top D_k^{-1}\mu_k = (5\%)^2/0.0925 = 0.027027028, \quad k = 1,\dots,3$$

Next, we compute:

$$D_k^{-1}\mu_k = 5\%/0.0925 = 0.54054054$$
$$s_k = \textstyle\prod_{j=k}^{N-1}(1+r_{f,j+1}) = 1.03^{3-k}, \quad k = 0,\dots,2$$
$$\beta_k = \textstyle\prod_{j=k}^{N-1}(1-\mu_{j+1}^\top D_{j+1}^{-1}\mu_{j+1}) = 0.972972973^{3-k}, \quad k = 0,\dots,2$$

The target expected final wealth is $z_0 = 1\times(1+0.06)^3$. (The problem says "grow the wealth by 6% in each period", implying $z_0 = (1.06)^3$).
The optimal strategy formula is $\pi_k^* = -A_k x_k + B_k$.
We calculate the coefficients:

$$A_k := (1+r_{f,k+1})D_{k+1}^{-1}\mu_{k+1} = 1.03\times 0.54054054 = 0.556756757, \quad k = 0,1,2$$
$$B_k := A_k s_k^{-1}\frac{z_0 - \beta_0 s_0 x_0}{1-\beta_0}$$

Substituting the values ($x_0 = 1$, $z_0 = (1.06)^3$, $s_0 = 1.03^3$, $\beta_0 = 0.97297^3$):

$$B_k = 0.556756757\times\frac{(1.06)^3 - 0.972972973^3\times 1.03^3\times 1}{1-0.972972973^3}\times 1.03^{k-3}$$
$$= 1.30187509\times 1.03^{k-3}$$

Thus, the values for B_k are:

$$B_k \approx \begin{cases} 1.91400139, & k = 0, \\ 1.227142143, & k = 1, \\ 1.263956407, & k = 2. \end{cases}$$

The optimal portfolio strategy is:

$$\pi_k^* = -0.556756757x_k + B_k, \quad k = 0,1,2$$